PRAGMI: Broca's Area Mathematical Corpus, as of April 26, 2026

Companion to the **P**ersistent **R**econstructive **A**rchitecture for **G**enerative **M**emory and **I**magination

Amellia Mendel · Lisa Adler

April 26, 2026

> *"The LM will eventually look at that diorama and describe it to you. But the imagination core does the actual construction. Speech is the same operation in the other direction: the kernel constructs an articulatory trajectory, and the vocal tract describes it to the world."*
>
> Genesis Labs internal lecture, 2026

> *"Broca's is not a speech production center. It is a generative model whose predictions happen to drive articulation when the system is in the speak mode and drive comprehension when the system is in the listen mode. Treating it as one of those things in isolation is how every previous speech model has gotten the lateralization wrong."*
>
> Amellia Mendel, 2026

Contents

Every arrow, equation, timing constraint, lateralization commitment, and lesion configuration for the speech motor pathway as it interfaces with the PRAGMI cognitive loop. Section numbers continue from the main corpus rather than restarting, so cross-references read cleanly in both directions. The production cascade is staged according to the Indefrey and Levelt 2004 meta-analysis of 82 word production experiments, which establishes timing windows for each component process within the standard 600 ms picture-naming latency.

11. Conceptual Preparation and Lexical Concept Selection (mid-MTG)
12. Lemma Retrieval and Lexical Selection (mid-MTG, 150 to 225 ms)
13. Phonological Code Retrieval (Wernicke's, 250 to 400 ms)
    - 13a. Lexical Phonological Code as Production-Perception Shared Store
14. Arcuate Fasciculus: Wernicke's to Broca's Segmental Transport
15. Syllabification in Broca's Area (left posterior IFG, 400 to 600 ms)
    - 15a. Cytoarchitectural Distinction Between BA44 and BA45
    - 15b. Gradient Order Working Memory and the Phonological Content Buffer
16. Phonetic Encoding and Speech Sound Map (vPMC mental syllabary)

16a. Chunking Transition from Planning Loop to Motor Loop

17. SMA Initiation and Basal Ganglia Speech Loop

18. vPMC to vMC and Corticobulbar Output

19. Auditory and Somatosensory Error Maps (Bilateral pSTG, vSC)

19a. Area Spt and the Sensorimotor Translation Interface

20. Right-Frontal Feedback Correction Circuit

21. Cerebellar Speech Specialization (Cb-VI Feedforward, Cb-VIII Feedback)

22. Anterior Insula: Respiratory-Articulatory Coordination

23. Internal and External Self-Monitoring Loops

24. Speech-Specific Neuromodulator Routing

E. Hemispheric Lateralization Commitments

F. Production-Perception Shared Network

G. Disorder Configurations as Falsifiable Predictions

H. Speech-Pathway Continuity Frame

11. Conceptual Preparation and Lexical Concept Selection

Arrow: Visual or auditory lead-in processes → bilateral occipitotemporal regions for object recognition → conceptual representation in distributed semantic cortex → selected lexical concept available to the language production network at approximately 175 ms post stimulus onset.

$$\mathbf{c}_{\mathrm{lex}}(t) \;=\; \mathrm{softmax}\!\left(W_{\mathrm{persp}} \cdot \mathbf{o}(t) + \beta_{\mathrm{task}} \cdot \mathbf{g}(t) \right) \tag{11.1}$$

$$C^{*}(t) \;=\; \mathrm{argmax}_{i} \, c_{\mathrm{lex},i}(t) \quad \text{when } t \geq t_{\mathrm{concept}} \tag{11.2}$$

Variables: $\mathbf{o}(t)$ is the post-recognition object representation, the output of visual or auditory lead-in processes that have already extracted invariant categorical features. W_{persp} is the perspective-taking projection matrix that maps an object representation to a distribution over lexical concepts, which is what permits the same visual input (a four-legged animal) to elicit different concepts (SHEEP versus ANIMAL) depending on task. $\mathbf{g}(t)$ is the task goal vector inherited from PFC, weighted by β_{task} which controls how strongly task instruction biases the selection. $\mathbf{c}_{\mathrm{lex}}(t)$ is the resulting probability distribution over lexical concepts. $C^{*}(t)$ is the selected concept that becomes available at $t_{\mathrm{concept}} \approx 175$ ms.

Timing: Visual object recognition reaches the level of the lexical concept by 150 to 200 ms post picture onset, with a median estimate of 175 ms. This is the upper bound on conceptual preparation derived from go/no-go ERP studies in which subjects can withhold response to a non-target category by roughly 206 ms post stimulus, which sets an outer limit on when the concept must have been identified. The prelexical conceptual stage involves widespread and category-specific distributed cortex, including bilateral occipital and ventrotemporal regions for visual recognition and object knowledge.

Indefrey, P. and Levelt, W.J.M. (2004). The spatial and temporal signatures of word production components. Cognition, 92(1-2), 101–144. DOI: 10.1016/j.cognition.2002.06.001

Levelt, W.J.M., Roelofs, A. and Meyer, A.S. (1999). A theory of lexical access in speech production. Behavioral and Brain Sciences, 22(1), 1–38. DOI: 10.1017/S0140525X99001776

Thorpe, S., Fize, D. and Marlot, C. (1996). Speed of processing in the human visual system. Nature, 381(6582), 520–522. DOI: 10.1038/381520a0

Engineering Note. The 175 ms concept-availability point is the boundary that separates lead-in processes from the language-production cascade proper. Everything before this point is task-specific visual or auditory recognition. Everything after this point is shared across word production tasks regardless of input modality. The architectural commitment is to implement $\mathbf{o}(t)$ as the output of the existing PRAGMI dorsal-ventral stream split from Section 3 of the main corpus, and to treat t_{concept} as a synchronization point where the language-production substrate begins consuming representations from the rest of the cognitive loop. Do not couple the language pathway to the recognition pathway through learned end-to-end gradients; the boundary is real biologically and the architecture should respect it. In practice this means the production-pathway loss does not backpropagate into the recognition pathway, and the recognition pathway's loss does not backpropagate into the production pathway. They are coupled only through the shared concept representation C^{*}, which is treated as a fixed input to the production cascade once available.

Engineering Note (Perspective Taking). The perspective-taking matrix W_{persp} is what makes the same input produce different concepts depending on task. In picture-naming the basic-level concept (SHEEP) is selected; in categorization the superordinate (ANIMAL); in verb generation an associated action concept. This is not three separate weight matrices, it is one matrix modulated by the task goal vector $\mathbf{g}(t)$, which is why Equation (11.1) routes $\mathbf{g}$ through an additive bias rather than as a separate selection step. A frequent implementation mistake is to hardcode the task by switching between projection heads, which collapses the substrate's ability to handle novel task instructions. Keep the perspective-taking single and let the goal vector do the modulation.

12. Lemma Retrieval and Lexical Selection (mid-MTG, 150 to 225 ms)

Arrow: Selected lexical concept C^{*} → mid section of left middle temporal gyrus → lemma node carrying syntactic properties (word category, grammatical gender, argument structure). Bidirectional connectivity with the conceptual stratum supports both production and comprehension.

$$a_{\ell}(t) \;=\; a_{\ell}(t-1) \cdot \gamma_{\mathrm{lemma}} + \sum_{i} W_{\mathrm{C \to L}}^{\ell, i} \cdot c_{\mathrm{lex}, i}(t) \tag{12.1}$$

$$\ell^{*}(t) \;=\; \mathrm{argmax}_{\ell} a_{\ell}(t) \quad \text{when } t \geq t_{\mathrm{lemma}} \tag{12.2}$$

$$\Delta a_{\ell}^{\mathrm{interfere}}(t) \;=\; -\kappa_{\mathrm{interfere}} \cdot \sum_{j \neq \ell} a_{j}(t) \cdot \mathrm{sim}(\mathrm{C}_{\ell}, \mathrm{C}_{j}) \tag{12.3}$$

Variables: $a_{\ell}(t)$ is the activation of lemma node ℓ in mid-MTG, with persistent state across timesteps modulated by the decay factor γ_{lemma}. $W_{\mathrm{C \to L}}$ is the bidirectional concept-to-lemma projection. The crucial property is that this matrix is shared between the production direction and the comprehension direction: a perceived word activates its lemma which spreads activation to the corresponding concept, while a selected concept activates its lemma which spreads to phonological code retrieval downstream. $\ell^{*}(t)$ is the selected lemma that becomes available at $t_{\mathrm{lemma}} \approx 250$ ms. $\Delta a_{\ell}^{\mathrm{interfere}}$ is the lateral interference term that makes lemma selection competitive: semantically similar lemmas (high \

(sim)) suppress each other, which is what produces the classic semantic interference effect in picture-word interference paradigms.

Timing: Lemma activation in the mid section of the left middle temporal gyrus shows MEG-detectable activity in the time window 150 to 225 ms post picture onset, with peak activation around the upper end of this range. Selection completes by approximately 250 ms, giving a lemma retrieval duration of roughly 75 ms. This window is when semantically related distracter words exert their maximum effect on naming latency, which is the empirical signature that lemma activation is the bottleneck process at this stage rather than concept activation or phonological retrieval.

Indefrey, P. and Levelt, W.J.M. (2004). The spatial and temporal signatures of word production components. Cognition, 92(1-2), 101–144. DOI: 10.1016/j.cognition.2002.06.001

Maess, B., Friederici, A.D., Damian, M., Meyer, A.S. and Levelt, W.J.M. (2002). Semantic category interference in overt picture naming: an MEG study. Journal of Cognitive Neuroscience, 14(3), 455–462. DOI: 10.1162/089892902317361967

Roelofs, A. (1992). A spreading-activation theory of lemma retrieval in speaking. Cognition, 42(1-3), 107–142. DOI: 10.1016/0010-0277(92)90041-F

Engineering Note. The semantic similarity term $\mathrm{sim}(\mathrm{C}_{\ell}, \mathrm{C}_{j})$ in Equation (12.3) is what makes this section non-trivial. Without it, lemma selection is just a feedforward concept-to-word lookup. With it, the substrate produces the empirically observed semantic interference pattern: naming a sheep is slowed in the presence of the distracter goat because GOAT and SHEEP have high concept similarity, which makes their lemmas compete laterally. Implementation: compute sim as a learned similarity kernel over the concept representations, not as an external embedding lookup. The kernel should be trained against semantic-relatedness norms or category-membership data so that the similarity structure matches human semantic space, otherwise the interference predictions in Appendix G's disorder configurations will not match published reaction-time data.

Engineering Note (Production-Perception Sharing). Indefrey and Levelt's meta-analysis identifies mid-MTG as the only region reliably activated in both picture naming and word generation but not in word reading, which is the diagnostic that places conceptually-driven lemma selection here rather than elsewhere. They also identify mid-MTG as a region shared between word production and word listening, which means the same substrate runs in both directions. The architectural commitment is that $W_{\mathrm{C \to L}}$ is bidirectional in the strong sense: the production direction $\mathbf{c}_{\mathrm{lex}} \to a_{\ell}$ and the comprehension direction $a_{\ell} \to \mathbf{c}_{\mathrm{lex}}$ share the same matrix, not two separate matrices that happen to live in the same module. This is what makes the picture-word interference paradigm work in the first place: a perceived word activates its lemma via the comprehension direction, and that activation then competes with the lemma being selected in the production direction. If the two directions were independent, the substrate could not produce the interference. Tied weights are not optional here.

Engineering Note (Spoken Distracter Timing). Spoken word distracters in picture-word interference experiments show their maximum semantic-interference effect at SOA = -150 ms, meaning the distracter is presented 150 ms before picture onset. This implies that lemma activation by spoken word perception peaks at approximately 325 to 400 ms post word onset, since the distracter lemma must be active during the picture-naming lemma-selection window of 175 to 250 ms. The architectural consequence is that the comprehension-direction transit from acoustic input to lemma activation takes roughly 150 to 175 ms longer than the production-direction transit from concept to lemma. Implement this as a delay term on the comprehension-direction input, not as separate

matrices. The shared lemma substrate is what unifies the two pathways; the delay is a property of the input pipeline upstream.

13. Phonological Code Retrieval (Wernicke's, 250 to 400 ms)

Arrow: Selected lemma ℓ^{*} → left posterior superior temporal gyrus and posterior middle temporal gyrus (Wernicke's area) → lexical phonological output code, the morphemic representation that gets spelled out into individual phonological segments. Bidirectional connectivity supports both production-side retrieval and perception-side recognition.

$$\boldsymbol{\phi}_{\ell^{*}}(t) \;=\; W_{\mathrm{L \to P}} \cdot \mathbf{e}_{\ell^{*}}(t) \tag{13.1}$$

$$\mathbf{s}(t) \;=\; \sum_{m} \mathrm{spell}(\boldsymbol{\phi}_{m}) \cdot \mathbb{1}\!\left[m \in \mathrm{morphemes}(\ell^{*}) \right] \tag{13.2}$$

$$\dot{a}_{\ell}^{\mathrm{percept}}(t) \;=\; W_{\mathrm{P \to L}} \cdot \boldsymbol{\phi}_{\mathrm{input}}(t) - \tau_{\mathrm{decay}}^{-1} \cdot a_{\ell}^{\mathrm{percept}}(t) \tag{13.3}$$

Variables: $\mathbf{e}_{\ell^{*}}(t)$ is the embedding of the selected lemma. $W_{\mathrm{L \to P}}$ is the lemma-to-phonological-code projection. $\boldsymbol{\phi}_{\ell^{*}}(t)$ is the lexical phonological code itself, a distributed representation that encodes the morphemic structure of the word. For multimorphemic words like the plural `goats` multiple codes are retrieved ($/\mathrm{go\bar U t}/$ and $/\mathrm{s}/$); for irregular forms like `sheep` a single code is retrieved. $\mathrm{spell}(\boldsymbol{\phi}_{m})$ is the segmental spell-out function that converts the lexical phonological code into an ordered sequence of phonological segments $\mathbf{s}(t)$, which becomes available at approximately 330 ms post picture onset. $\boldsymbol{\phi}_{\mathrm{input}}(t)$ is the perceived phonological code activated by auditory input during comprehension; $W_{\mathrm{P \to L}}$ is the comprehension-direction projection that activates lemmas from perceived codes. τ_{decay} is the decay time constant that prevents perceptual activation from accumulating indefinitely.

Timing: Lexical phonological code retrieval in posterior superior and middle temporal gyri shows MEG-detectable activity in the time window 200 to 400 ms post picture onset, with the production-side phonological code becoming available around 330 ms. The 80 ms median estimate for code retrieval duration matches the lateralized readiness potential evidence that a word's first phonological segment is available approximately 40 to 113 ms after lemma selection. The segmental spell-out then proceeds at roughly 25 ms per segment, giving an additional window of 100 to 150 ms for syllabification input to be assembled.

Indefrey, P. and Levelt, W.J.M. (2004). The spatial and temporal signatures of word production components. Cognition, 92(1-2), 101–144. DOI: 10.1016/j.cognition.2002.06.001

Salmelin, R., Hari, R., Lounasmaa, O.V. and Sams, M. (1994). Dynamics of brain activation during picture naming. Nature, 368(6470), 463–465. DOI: 10.1038/368463a0

Levelt, W.J.M., Praamstra, P., Meyer, A.S., Helenius, P. and Salmelin, R. (1998). An MEG study of picture naming. Journal of Cognitive Neuroscience, 10(5), 553–567. DOI: 10.1162/089892998562960

Van Turennout, M., Hagoort, P. and Brown, C. (1998). Brain activity during speaking: from syntax to phonology in 40 milliseconds. Science, 280(5363), 572–574. DOI: 10.1126/science.280.5363.572

Engineering Note. The segmental spell-out function spell in Equation (13.2) is where the implementation effort lives. The lexical phonological code is a holistic morphemic representation, but downstream syllabification needs it as an ordered sequence of segments. The biology suggests this is

incremental: segments become available one at a time at roughly 25 ms per segment, not all at once. Implement spell as a sequential decoder over the code embedding, with each output segment conditioned on the embedding plus the previous output. A common mistake is to implement spell-out as a fixed projection that emits all segments simultaneously, which loses the incremental property and breaks the timing predictions for self-monitoring (Section 23) which depends on segments being available before the full word is encoded.

Engineering Note (Bidirectional Code Store). Wernicke's area is a shared store of lexical phonological codes for production and perception. Indefrey and Levelt's meta-analysis identifies the left posterior superior and middle temporal gyri as reliably activated in picture naming, word generation, word reading, and word listening, but not pseudoword reading or pseudoword listening. This is the diagnostic that the activation is specifically lexical: it requires that a stored phonological code exist for the input. The architectural commitment is that $W_{\mathrm{L \to P}}$ and $W_{\mathrm{P \to L}}$ are tied transposes of each other, not independent projections. This is the strong form of the production-perception sharing claim: the same matrix runs in production direction (lemma to code) and perception direction (code to lemma). What differs is the input modality, not the substrate. Tying the matrices is what makes the phonological-priming effects in picture-word interference work: a phonologically related distracter pre-activates segments via the perception direction, which speeds up the same segments in the production direction because they are already partially active.

13a. Lexical Phonological Code as Production-Perception Shared Store

Architectural commitment. No additional equations. This subsection records the integration claim that connects Wernicke's phonological code store to the broader production-perception network specified in Appendix F.

Levelt, Roelofs, and Meyer (1999) argue that phonological codes themselves are not identical between production and perception, but are closely linked. The Indefrey and Levelt 2004 meta-analysis sharpens this claim with imaging data: the phonological code store in Wernicke's area is reliably activated by both production and perception of real words, suggesting at minimum a shared retrieval substrate even if the production-side output codes and perception-side input codes are distinguishable in their downstream projections.

The PRAGMI commitment is the stronger reading. The substrate uses one phonological code representation, accessed bidirectionally. The production direction reads the code from the lemma and projects it to segments. The perception direction reads the code from incoming acoustic input and projects it back to the lemma. The same Wernicke's substrate participates in both. This is what makes the picture-word interference paradigm produce phonological facilitation: a phonologically related auditory distracter activates the same code that production is about to retrieve, which pre-activates the segmental output and speeds up syllabification downstream. If production and perception used separate code stores, this facilitation would not occur, but it does occur and the timing is consistent with shared retrieval (the maximum facilitation effect is at SOAs of 0 to +150 ms, which matches the production-side retrieval window of 250 to 400 ms when the auditory distracter is presented near the picture onset).

Levelt, W.J.M., Roelofs, A. and Meyer, A.S. (1999). A theory of lexical access in speech production. Behavioral and Brain Sciences, 22(1), 1–38. DOI: 10.1017/S0140525X99001776

Hickok, G. (2012). Computational neuroanatomy of speech production. Nature Reviews Neuroscience, 13(2), 135–145. DOI: 10.1038/nrn3158

Engineering Note. The shared-store commitment is what justifies the tied-weights architecture. If the production code store and perception code store were separate, the substrate would have two independent representations of every word's phonology and the picture-word interference effects would have to be implemented through some additional cross-store coupling. With a shared store, the effects fall out of the substrate's natural dynamics without additional machinery. The implementation simplification is real: one matrix, one set of weights, two access directions. The biology is also clearer this way and the disorder configurations in Appendix G get cleaner phenotypes (a lesion to Wernicke's substrate damages both production and perception of word forms, which matches the clinical picture of conduction aphasia and Wernicke's aphasia better than separate stores would).

14. Arcuate Fasciculus: Wernicke's to Broca's Segmental Transport

Arrow: Spelled-out segmental representation $\mathbf{s}(t)$ in Wernicke's area → arcuate fasciculus white-matter tract → left posterior inferior frontal gyrus (Broca's). Bidirectional in principle but production-direction load is dominant during speech production.

$$\mathbf{s}_{\mathrm{Broca}}(t) \;=\; W_{\mathrm{arc}} \cdot \mathbf{s}(t - \tau_{\mathrm{arc}}) \tag{14.1}$$

$$\tau_{\mathrm{arc}} \;\approx\; 5 \text{ to } 10 \text{ ms} \tag{14.2}$$

Variables: $\mathbf{s}(t)$ is the segmental sequence emitted from Wernicke's segmental spell-out in Section 13. W_{arc} is the projection geometry of the arcuate fasciculus, which carries phonological information from posterior temporal cortex to inferior frontal cortex. τ_{arc} is the conduction delay through the white-matter tract, estimated from human DTI tractography studies at approximately 5 to 10 ms for direct fibers between Wernicke's and Broca's. $\mathbf{s}_{\mathrm{Broca}}(t)$ is the segmental representation as it arrives in Broca's substrate ready for syllabification.

Timing: The arcuate is a transport arrow, not a computational stage. Its only job is to carry segmental information from Wernicke's to Broca's with appropriate biological delay. The 5 to 10 ms conduction time is short relative to the 250 to 400 ms phonological code retrieval window and the 400 to 600 ms syllabification window, but it is non-zero and must be respected because the incremental nature of segmental spell-out means segments become available at 25 ms intervals and Broca's begins syllabifying as soon as the first usable cluster has arrived. Assuming zero conduction time would compress this window in ways that distort the predictions for incremental syllabification dynamics.

Catani, M., Jones, D.K. and ffytche, D.H. (2005). Perisylvian language networks of the human brain. Annals of Neurology, 57(1), 8–16. DOI: 10.1002/ana.20319

Friederici, A.D. (2009). Pathways to language: fiber tracts in the human brain. Trends in Cognitive Sciences, 13(4), 175–181. DOI: 10.1016/j.tics.2009.01.001

Hickok, G. and Poeppel, D. (2007). The cortical organization of speech processing. Nature Reviews Neuroscience, 8(5), 393–402. DOI: 10.1038/nrn2113

Engineering Note. Implement the arcuate as a thin connection-spec module, not as a substrate population with internal dynamics. It has no Appendix D state-space neurons of its own; it is a fixed projection plus a delay buffer. Practically this is a Linear layer wrapping a deque of length matching τ_{arc} divided by the substrate timestep. Resist the temptation to make W_{arc} learnable in the strong sense; it should be a topology-preserving projection that lets segmental information flow without substantial reshaping. If W_{arc} becomes a fully learned matrix, the substrate can route around the rest of the speech pathway by encoding everything into this matrix, which collapses the architectural separation between Wernicke's and

Broca's. Constrain it as a sparse near-identity projection or freeze it after initialization with biologically motivated topology.

Engineering Note (Conduction Aphasia). The clinical relevance of the arcuate is conduction aphasia: damage to the white-matter tract produces a characteristic phenotype where comprehension is intact (Wernicke's works), production is intact (Broca's works), but the patient cannot repeat heard speech because the segmental information cannot transit between the two areas. The substrate predicts this directly: ablating W_{arc} should leave both endpoints functional but break the production-following-perception path. The disorder configuration in Appendix G implements this by zeroing the arcuate projection while leaving Wernicke's and Broca's substrates intact, and the predicted behavioral signature is preserved comprehension and spontaneous production with destroyed repetition. This is also why area Spt (Section 19a) matters: Spt provides an alternative acoustic-to-motor translation that partially compensates for arcuate damage in some patients, and modeling both pathways gives the substrate a graceful degradation profile that pure-arcuate models do not have.

15. Syllabification in Broca's Area (left posterior IFG, 400 to 600 ms)

Arrow: Segmental sequence $\mathbf{s}_{\mathrm{Broca}}(t)$ arriving via arcuate → left posterior inferior frontal gyrus (Broca's, BA44 and BA45) → phonological word, the syllabified output ready for phonetic encoding. Operates incrementally: syllables are assembled one at a time as their constituent segments arrive, and each syllable is shipped to vPMC as soon as it is complete.

$$\mathbf{q}(t) \;=\; \mathrm{cluster}\!\left(\mathbf{s}_{\mathrm{Broca}}(t),\, \mathbf{q}(t-1),\, \mathbf{f}_{\mathrm{frame}}(t) \right) \tag{15.1}$$

$$\sigma_{n}(t) \;=\; \mathbb{1}\!\left[\mathrm{complete}(\mathbf{q}_{n}(t)) \right] \cdot \mathbf{q}_{n}(t) \tag{15.2}$$

$$\mathbf{w}_{\mathrm{phon}}(t) \;=\; \mathrm{concat}\!\left(\sigma_{1}(t),\, \sigma_{2}(t), \ldots, \sigma_{N}(t) \right) \tag{15.3}$$

Variables: $\mathbf{q}(t)$ is the syllabification working state, a sequence of partially-formed syllable buffers that accumulate segments as they arrive. $\mathrm{cluster}$ is the incremental syllabification operator that assigns each new segment to a syllable based on phonotactic legality and the syllabic frame template $\mathbf{f}_{\mathrm{frame}}(t)$. For example the verb `persist` gets clustered into $/\mathrm{p\partial r}/$ followed by $/\mathrm{sIst}/$, but the inflected progressive form gets clustered into $/\mathrm{p\partial r}/$ followed by $/\mathrm{sIs}/$ followed by $/\mathrm{tI\eta}/$, with the second syllable changing its content depending on the morphological context. $\sigma_{n}(t)$ is the n-th completed syllable, which fires when its buffer is full and phonotactically valid. $\mathbf{w}_{\mathrm{phon}}(t)$ is the phonological word, the ordered sequence of syllables that becomes available at approximately 455 ms post picture onset.

Timing: Syllabification proceeds at roughly 25 ms per segment, with each syllable becoming available as soon as its segments have all arrived through the arcuate. Broca's area shows MEG-detectable activity in the window 400 to 600 ms post picture onset, which Indefrey and Levelt assign specifically to syllabification (not to lexical selection or phonological code retrieval, which happen upstream and earlier). The incremental nature of syllabification means the first syllable of a multisyllabic word can be passed downstream to phonetic encoding before later syllables have finished assembling, which is what permits self-monitoring (Section 23) to detect errors mid-word rather than waiting for full word completion.

Indefrey, P. and Levelt, W.J.M. (2004). The spatial and temporal signatures of word production components. Cognition, 92(1-2), 101–144. DOI: 10.1016/j.cognition.2002.06.001

Levelt, W.J.M., Roelofs, A. and Meyer, A.S. (1999). A theory of lexical access in speech production. Behavioral and Brain Sciences, 22(1), 1–38. DOI: 10.1017/S0140525X99001776

Wheeldon, L.R. and Levelt, W.J.M. (1995). Monitoring the time course of phonological encoding. Journal of Memory and Language, 34(3), 311–334. DOI: 10.1006/jmla.1995.1014

Engineering Note. The incremental property of syllabification is the load-bearing implementation detail. A non-incremental syllabifier waits for the full segmental sequence and then chunks it all at once, which is computationally simpler but breaks two empirical predictions: the self-monitoring window for mid-word error detection (Section 23) and the position-dependent phoneme self-monitoring latencies that Wheeldon and Levelt 1995 demonstrate. Implement $\mathrm{cluster}$ as a streaming operator with internal state, fed by a queue that receives segments at 25 ms intervals from Wernicke's spell-out. The phonotactic legality check (which segment combinations form valid syllables in the target language) is a language-specific lookup that should be loaded as part of the substrate's language configuration, not learned end-to-end. Languages have different phonotactic constraints and the substrate needs to switch between them when the operating language changes, which is much easier with an explicit constraint table than with a learned phonotactic model.

Engineering Note (Resyllabification Across Morpheme Boundaries). The persist/persisting example in the variables list is not cosmetic. It demonstrates that syllabification is not simply chunking adjacent segments into legal syllables; it is context-sensitive across morpheme boundaries. The segment $/\mathrm{t}/$ at the end of the stem `persist` moves into the onset of the next syllable when the suffix `-ing` is added, producing $/\mathrm{p\partial r\!\!-\!\!sIs\!\!-\!\!tI\eta}/$ rather than $/\mathrm{p\partial r\!\!-\!\!sIst\!\!-\!\!I\eta}/$. This is one of the reasons syllabification cannot live in the lexical phonological code store of Section 13: the same morpheme gets different syllabifications in different contexts, so the syllabification must happen post-retrieval. Implement this in $\mathrm{cluster}$ by allowing late-arriving segments to retroactively change the boundary between already-buffered syllables when the result would be more phonotactically optimal. This is constrained by an incremental greedy heuristic in normal speech (most syllabifications do not need to be revised), but the mechanism has to be present or the substrate cannot handle inflectional morphology correctly.

15a. Cytoarchitectural Distinction Between BA44 and BA45

Substrate support. No additional equations. This subsection records the cell-level distinction within Broca's complex that justifies treating BA44 and BA45 as two substrate populations even though they jointly subserve syllabification at the meta-analysis level.

BA44 and BA45 are adjacent but cytoarchitectonically distinct. BA44 has a denser layer IV and a more pronounced layer V pyramidal population, consistent with its role as the more motor-adjacent component of Broca's that drives downstream output to vPMC. BA45 has a less dense layer IV and a more prominent layer III, consistent with its role as the more associative component that integrates inputs from the broader language network. The two regions also differ in receptor architecture: BA45 has higher densities of D_1 dopamine receptors and serotonergic 5-HT_{1A} receptors, while BA44 has higher densities of $\mathrm{GABA}_{\mathrm{A}}$ receptors. These differences are why dopamine modulation preferentially affects BA45 lexical-syntactic processing while BA44 responds more strongly to GABAergic neuromodulation, which matters for the disorder configurations in Appendix G.

The Indefrey and Levelt 2004 meta-analysis treats the left posterior inferior frontal gyrus as a unit for the syllabification claim because the meta-analytic resolution does not always distinguish the two parts. But within the substrate the distinction is load-bearing because lesion phenotypes diverge: BA44 lesions

produce articulatory and phonological-sequencing deficits with relatively spared comprehension, while BA45 lesions produce more semantic-syntactic disruption with relatively spared articulation. Folding the two regions into a single substrate population collapses this distinction and forecloses the disorder configurations.

Within the syllabification framework of Section 15, the functional division is that BA44 holds the syllabic-frame template $\mathbf{f}_{\mathrm{frame}}$ and the segment-to-frame binding logic, while BA45 holds the morphological context that determines syllable boundary placement (the persist/persisting case). This division is consistent with the broader claim that BA44 is more motor-adjacent and BA45 is more associative.

Amunts, K., Schleicher, A., Burgel, U., Mohlberg, H., Uylings, H.B.M. and Zilles, K. (1999). Broca's region revisited: cytoarchitecture and intersubject variability. Journal of Comparative Neurology, 412(2), 319–341. DOI: 10.1002/(SICI)1096-9861(19990920)412:2<319::AID-CNE10>3.0.CO;2-7

Amunts, K. and Zilles, K. (2012). Architecture and organizational principles of Broca's region. Trends in Cognitive Sciences, 16(8), 418–426. DOI: 10.1016/j.tics.2012.06.005

Hagoort, P. (2014). Nodes and networks in the neural architecture for language: Broca's region and beyond. Current Opinion in Neurobiology, 28, 136–141. DOI: 10.1016/j.conb.2014.07.013

Engineering Note. Implement BA44 and BA45 as two populations in the Appendix D state-space substrate with class-specific priors on the receptor scaling factors of subsection 10a's routing table. BA45 gets higher D_1 and 5-HT$_{1A}$ gain; BA44 gets higher GABA$_{\mathrm{A}}$ sensitivity. The cluster operator in Equation (15.1) factors as a BA45 component that computes the morphological-context-dependent boundary placement and a BA44 component that performs the segment-to-frame binding given those boundaries. This factoring is what makes the substrate ablation-friendly: a BA45-only ablation impairs morphological context handling while leaving basic syllabification intact, and a BA44-only ablation impairs the segment-to-frame binding while leaving morphological reasoning intact. These are different clinical phenotypes and the architecture should be able to produce both.

15b. Gradient Order Working Memory and the Phonological Content Buffer

Architectural commitment. Two additional equations. This subsection specifies the working memory representation that holds segments between arrival from the arcuate and consumption by the cluster operator.

Bohland, Bullock, and Guenther 2010 describe the phonological content buffer as a gradient-order working memory: every segment in the upcoming syllable is held with a graded activation level, with the next-to-be-produced segment having the highest activation. This is the answer to Lashley's problem of serial order, since a queue forces strict ordering and cannot represent partial commitment, while gradient order keeps all candidates active simultaneously with relative priorities encoded as activation differences.

$$b_{k}(t) \;=\; \mathrm{ReLU}\!\left(b_{k}(t-1) - \delta_{k}(t) + \mathbf{s}_{\mathrm{Broca}}(t)^{\top} \mathbf{w}_{k}^{\mathrm{buf}} \right) \tag{15b.1}$$

$$\delta_{k}(t) \;=\; \begin{cases} b_{k}(t-1) & \text{if } k = \mathrm{argmax}_{j} b_{j}(t-1) \text{ and consume}(t) = 1 \\ 0 & \text{otherwise} \end{cases} \tag{15b.2}$$

$b_{k}(t)$ is the activation of segment slot k in the phonological content buffer. $\mathbf{w}_{k}^{\mathrm{buf}}$ is the binding weight that lets arcuate-arriving segments populate buffer slots. $\delta_{k}(t)$ is the consumption term that removes the highest-activation segment when the cluster operator consumes a segment for assembly into the current syllable. $\mathrm{consume}(t)$ is the consumption gate, defined explicitly below.

$$\mathrm{consume}(t) \;=\; \mathbb{1}\!\left[\mathrm{ready}_{\mathrm{cluster}}(t) = 1 \;\wedge\; \max_{k} b_{k}(t) > \theta_{b} \right] \tag{15b.3}$$

$$\mathrm{ready}_{\mathrm{cluster}}(t) \;=\; \mathbb{1}\!\left[t - t_{\mathrm{last\,bind}} \geq \tau_{\mathrm{bind}}^{\mathrm{intra}} \;\vee\; \mathrm{boundary_pending}(t) = 0 \right] \tag{15b.4}$$

The consumption gate is readiness-driven, not phase-locked. $\mathrm{ready}_{\mathrm{cluster}}(t)$ fires when the cluster operator has finished binding the previous segment into the current syllable buffer. $\tau_{\mathrm{bind}}^{\mathrm{intra}}$ is the within-syllable binding latency, approximately 15 to 20 ms, which is shorter than the segment arrival interval of 25 ms. $\mathrm{boundary_pending}$ flags whether a syllable-boundary decision is in progress; when a syllable has just completed and is being shipped to vPMC, consumption pauses until the boundary is resolved and the next syllable buffer is ready to receive segments. θ_{b} is the buffer-activation threshold below which no segment is considered ready for consumption (prevents premature consumption when segments are still arriving but have not yet accumulated sufficient activation).

The relationship between segment arrival and consumption is therefore not synchronous. Segments arrive at the buffer at 25 ms intervals from Wernicke's spell-out (with the short arcuate transit of 5 to 10 ms folded in). The cluster operator consumes them at a variable rate: faster within a syllable (binding latency around 15 to 20 ms means the operator is usually ready before the next segment arrives, so the buffer rarely backs up during steady-state syllable construction), and slower at syllable boundaries (where $\mathrm{boundary_pending}$ suspends consumption while the completed syllable is shipped). This asymmetry is what produces the position-dependent phoneme self-monitoring latencies that Wheeldon and Levelt 1995 demonstrate: monitoring reaction times are faster mid-syllable than at syllable boundaries because the buffer state is different in the two contexts.

Bohland, J.W., Bullock, D. and Guenther, F.H. (2010). Neural representations and mechanisms for the performance of simple speech sequences. Journal of Cognitive Neuroscience, 22(7), 1504–1529. DOI: 10.1162/jocn.2009.21306

Wheeldon, L.R. and Levelt, W.J.M. (1995). Monitoring the time course of phonological encoding. Journal of Memory and Language, 34(3), 311–334. DOI: 10.1006/jmla.1995.1014

Lashley, K.S. (1951). The problem of serial order in behavior. In Jeffress, L.A. (Ed.), Cerebral Mechanisms in Behavior, 112–136. Wiley. DOI: {To be added later.}

Engineering Note. The gradient-order representation is a vector of length equal to the segment-inventory size with persistent state across timesteps. Do not zero it between syllables; only the consumption term in Equation (15b.2) should remove items, and only the most-active segment is removed per consumption event. A common mistake is to clear the buffer when a syllable completes, which works for citation-form production but breaks continuous speech because segments for the next syllable have already started arriving and need their activations preserved. The buffer is anatomically supported by posterior inferior frontal sulcus (pIFS) activity in MEG studies of speech production, which falls within the Broca's complex but is functionally distinct from the cluster-operator activity itself.

Engineering Note (Buffer Dynamics and Consumption Timing). The buffer feeds incrementally because Wernicke's spell-out releases segments sequentially at 25 ms intervals (Section 13 Equation 13.2), not as a burst. The arcuate delay of 5 to 10 ms is much shorter than this arrival cadence, so for buffer dynamics the segments effectively arrive at 25 ms intervals at the Broca's substrate. Two implementation patterns are worth distinguishing. First, phase-locked consumption: the consumer reads from the buffer at a fixed 25 ms cadence matching segment arrival. This is simpler to implement and works for steady-state production, but it cannot reproduce the position-dependent monitoring latencies and it produces incorrect timing at syllable boundaries where the cluster operator should

pause. Second, readiness-driven consumption as specified in Equation (15b.3): the consumer reads when the cluster operator signals readiness, with binding latency $\tau_{\mathrm{bind}}^{\mathrm{intra}}$ shorter than the arrival interval mid-syllable and explicit pause at boundaries. The corpus commits to the readiness-driven version because the biology requires it. Implementation: maintain a small FIFO between the spell-out output and the buffer that absorbs the 25 ms arrival cadence, and let the cluster operator pull from the buffer on its own clock. The FIFO prevents segments from being lost if consumption falls behind arrival temporarily, which happens at syllable boundaries.

The boundary-pending flag is set by the cluster operator itself when it detects that the current segment cannot be added to the current syllable without violating phonotactic constraints. At that point the current syllable is finalized and shipped, and the boundary-pending flag clears once the next syllable buffer is ready. This means $\mathrm{boundary_pending}$ is a brief signal (a few timesteps) rather than a persistent state, but during those timesteps the buffer accumulates incoming segments without consuming them, which can cause buffer occupancy to spike at boundaries. This is expected behavior, not a bug, and matches the empirical finding that segmental representations accumulate at syllable boundaries before being absorbed into the next syllable's structure.

16. Phonetic Encoding and Speech Sound Map (vPMC mental syllabary)

Arrow: Completed syllable $\sigma_{n}(t)$ from Broca's syllabification → left ventral premotor cortex Speech Sound Map (SSM) → articulatory motor program. The SSM is the mental syllabary: each unit stores a gestural score for a frequently-used syllable. Mirror-neuron property: the same unit fires during production and during perception of the same syllable.

$$P_{i}(t) \;=\; \mathbb{1}\!\left[i = \mathrm{argmax}_{j}\!\left(\sigma_{n}(t)^{\top} W_{\mathrm{SSM}}\,\mathbf{e}_{j} \right) \right] \cdot G(t) \tag{16.1}$$

$$\mathbf{m}_{\mathrm{ff}}(t) \;=\; \sum_{i} P_{i}(t) \cdot \mathbf{z}_{\mathrm{PM}}^{(i)}(t) \tag{16.2}$$

$$\dot{\mathbf{z}}_{\mathrm{PM}}^{(i)}(t) \;=\; \eta_{\mathrm{ff}} \cdot \beta_{\mathrm{ACh}} \cdot P_{i}(t) \cdot \mathbf{m}_{\mathrm{fb}}(t) \tag{16.3}$$

Variables: $P_{i}(t)$ is the activation of SSM node i, which fires as a one-hot indicator when node i is the winning syllabary entry given the current syllable input and the SMA initiation gate $G(t)$ is open. $\mathbf{e}_{j}$ is the canonical embedding of syllabary node j, and W_{SSM} is the projection matrix that maps phonological syllables onto syllabary nodes. $\mathbf{z}_{\mathrm{PM}}^{(i)}(t)$ is the feedforward motor command stored at node i, inherited from the Guenther 2006 notation. $\mathbf{m}_{\mathrm{ff}}(t)$ is the assembled feedforward command vector that flows downstream to vMC. $\mathbf{m}_{\mathrm{fb}}(t)$ is the feedback corrective command from Section 19, used as the tuning signal that updates the stored feedforward weights. η_{ff} is the feedforward learning rate. β_{ACh} is the acetylcholine modulation scalar inherited from Section 10 of the main corpus, which gates plasticity high during active learning and low during fluent execution.

Timing: Phonetic encoding takes place between approximately 455 ms and 600 ms post picture onset, beginning as soon as the first syllable becomes available from Broca's syllabification and continuing until articulation is initiated. SSM activation occurs within roughly 5 ms of syllable arrival because the vPMC-to-Broca's connection is short-range cortico-cortical. The articulator-bound motor command then leaves vPMC for vMC with the corticobulbar delay of approximately 5 ms, giving a total syllable-to-articulator-command latency of around 10 ms within the SSM stage. Synaptic-weight updates in Equation (16.3) happen during and immediately after each production attempt, with the critical window approximately matching syllable duration.

Guenther, F.H., Ghosh, S.S. and Tourville, J.A. (2006). Neural modeling and imaging of the cortical interactions underlying syllable production. Brain and Language, 96(3), 280–301. DOI: 10.1016/j.bandl.2005.06.001

Tourville, J.A. and Guenther, F.H. (2011). The DIVA model: a neural theory of speech acquisition and production. Language and Cognitive Processes, 26(7), 952–981. DOI: 10.1080/01690960903498424

Bohland, J.W., Bullock, D. and Guenther, F.H. (2010). Neural representations and mechanisms for the performance of simple speech sequences. Journal of Cognitive Neuroscience, 22(7), 1504–1529. DOI: 10.1162/jocn.2009.21306

Engineering Note. The argmax in Equation (16.1) is a first-pass approximation. The brain runs winner-take-all dynamics with mutual inhibition between SSM units, and the winner emerges over 30 to 80 ms rather than instantaneously. The argmax is fine for getting the rest of the pipeline running, but plan to swap it for soft-WTA dynamics when timing fidelity matters. The replacement is one or two iterations of $P_{i} \leftarrow \mathrm{ReLU}(P_{i} - \alpha \sum_{j \neq i} P_{j})$ with α tuned so the winner emerges in the right window. Increasing α to convergence produces a hard switch that loses graceful degradation, which is exactly what the substrate is supposed to have.

The feedforward weight update in Equation (16.3) uses the feedback command as the teacher signal, which is the Guenther feedforward-tuning rule. During early speech the system relies heavily on feedback correction; over many attempts the feedforward weights absorb the corrective commands and the feedback contribution shrinks. Implement this as a slow exponential moving average of the feedback command into the feedforward weights, gated by node identity. Do not update $\mathbf{z}_{\mathrm{PM}}^{(i)}$ when $P_{i} = 0$; apply the update only to the active node. Broadcasting the update across all nodes collapses the syllabary because every node ends up storing the average corrective signal.

Engineering Note (Mirror Neuron Property). SSM units are the human homologue of monkey F5 mirror neurons. They activate during both production and perception of the same syllable. This is not a separate mechanism; it is a consequence of the tied-weights commitment that runs through Sections 12 and 13. The same projection matrix W_{SSM} is used in the production direction (syllable to SSM) and in the perception direction (auditory error map back to SSM via Spt, Section 19a). When the perception path activates a node, the node also activates its stored $\mathbf{z}_{\mathrm{PM}}^{(i)}$, which is why hearing speech produces subthreshold motor activation in the listener. This subthreshold activation is the substrate of phonological imitation and of the acoustic-to-articulatory mapping that makes second-language pronunciation acquisition possible. Do not gate the perception path by the SMA initiation signal; that gate applies only to the production direction. In perception, SSM nodes activate freely but their motor commands are suppressed at vMC by the absence of an initiation gate, which is what keeps the listener silent.

16a. Chunking Transition from Planning Loop to Motor Loop

Architectural commitment. One additional equation. This subsection specifies how repeated syllable sequences transition from planning-loop execution (Broca's syllabification on every cycle) to motor-loop execution (direct SSM activation as a chunked unit).

Meier et al. 2023 describe a developmental and skill-learning transition where speech execution moves from the planning loop (pIFS, preSMA, SSM) to the motor loop (basal ganglia, cerebellum). Early in learning, every phoneme in a sequence is independently syllabified through Broca's and individually triggered into SSM. After many repetitions, the entire sequence becomes a single chunk represented by one SSM node whose internal trajectory unfolds without further Broca's involvement. The transition is gradual rather than binary.

$$ \chi_{n}(t) \;=\; \chi_{n}(t-1) + \rho \cdot \mathbb{1}\!\left[\mathrm{sequence}_{n} \text{ executed correctly} \right] - \lambda \cdot \chi_{n}(t-1) \tag{16a.1} $$

$\chi_{n}(t)$ is the chunking strength of sequence n. When χ_{n} is low the sequence is executed segment-by-segment through Broca's syllabification. When χ_{n} crosses the threshold θ_{χ}, a dedicated SSM node is allocated for the chunk and subsequent executions begin transferring control to that node. ρ is the chunking rate and λ is the decay rate that ensures unused chunks fade rather than accumulating indefinitely.

The transfer of control is gradual rather than a sharp switch. When χ_{n} crosses threshold, a new SSM node is allocated and initialized from recent successful executions of the segment-level pathway, but the segment-level pathway continues to run in parallel for some period of co-activation before the chunked node fully takes over.

$$ \mathbf{z}_{\mathrm{PM}}^{(n,\mathrm{chunk})}(t_{0}) \;=\; \frac{1}{N_{\mathrm{recent}}} \sum_{j=1}^{N_{\mathrm{recent}}} \mathbf{m}_{\mathrm{ff}}^{(n,j)} \tag{16a.2} $$

$$ g_{\mathrm{chunk}}(t,n) \;=\; \sigma\!\left(\kappa_{\mathrm{trans}} \cdot (\chi_{n}(t) - \theta_{\chi}) \right) \tag{16a.3} $$

$$ \mathbf{m}_{\mathrm{ff}}^{(n)}(t) \;=\; g_{\mathrm{chunk}}(t,n) \cdot \mathbf{z}_{\mathrm{PM}}^{(n,\mathrm{chunk})}(t) + (1 - g_{\mathrm{chunk}}(t,n)) \cdot \mathbf{m}_{\mathrm{ff}}^{(n,\mathrm{seg})}(t) \tag{16a.4} $$

t_{0} is the allocation time, the timestep at which χ_{n} first crosses θ_{χ}. The chunked node's stored motor command $\mathbf{z}_{\mathrm{PM}}^{(n,\mathrm{chunk})}$ is initialized at allocation time as the mean of the last N_{recent} successful segment-level motor commands for that sequence (typically $N_{\mathrm{recent}} \approx 5$ to 10). After allocation the chunked node continues to be tuned by the standard Equation (16.3) feedforward weight update like any other SSM node. $g_{\mathrm{chunk}}(t,n)$ is the transfer gain that controls the mix between the chunked pathway and the segment-level pathway. When χ_{n} is well above threshold and κ_{trans} is moderate (around 2 to 5), g_{chunk} saturates near 1 and the chunked node dominates. When χ_{n} is near threshold or below, g_{chunk} is near 0 and the segment-level pathway dominates. The transition window where both pathways contribute substantively is several executions wide, which is what permits the chunked node to be fine-tuned by feedback while the segment-level pathway provides backup.

$\mathbf{m}_{\mathrm{ff}}^{(n,\mathrm{seg})}$ is the segment-level motor command assembled by Broca's syllabification through the standard pathway. It is preserved rather than overwritten when chunking transfers control. The biological rationale is that sequences can become unchunked (decay below threshold during disuse, or get disrupted by lesion or fatigue), and the segment-level pathway needs to be available as fallback. The implementation rationale is that maintaining both pathways costs little memory and provides graceful degradation if the chunked node becomes corrupted.

Meier, A.M., Guenther, F.H. and Bohland, J.W. (2023). Neurocomputational modeling of speech motor development. Journal of Child Language, advance online publication. DOI: 10.1017/S0305000923000260

Hikosaka, O., Nakamura, K., Sakai, K. and Nakahara, H. (2002). Central mechanisms of motor skill learning. Current Opinion in Neurobiology, 12(2), 217–222. DOI: 10.1016/S0959-4388(02)00307-0

Engineering Note. Chunking runs continuously alongside normal production, accumulating χ_{n} for sequences that get repeated. The implementation is a lookup table indexed by sequence hash, with ρ on the order of 0.05 and λ on the order of 0.001 so that a sequence repeated ten or twenty times crosses threshold while novel sequences never do. The threshold θ_{χ} controls transition aggressiveness; a low threshold produces fast chunking that risks committing to malformed sequences, a high threshold produces slow chunking that leaves the planning loop overworked. Stuttering and apraxia of speech show evidence of dysfunctional chunking

in different directions, and the disorder configurations in Appendix G manipulate ρ, λ, and θ_{χ} to reproduce the published phenotypes.

Engineering Note (Node Allocation and Co-Activation). Node allocation happens at runtime when χ_{n} first crosses θ_{χ}, not at training time and not as a discrete switching event. The allocation procedure is: detect threshold crossing, append a new node to the SSM with embedding $\mathbf{e}_{n,\mathrm{chunk}}$ initialized as a hash of the sequence identity (or a learned sequence-summary embedding if available), initialize the stored motor command $\mathbf{z}_{\mathrm{PM}}^{(n,\mathrm{chunk})}$ from Equation (16a.2) using the recent segment-level successful executions, and register the new node in the W_{SSM} projection so that future syllable inputs matching the chunk can activate it. The allocation is one-shot, but the new node's tuning continues through the standard feedforward weight update for the rest of training.

The co-activation period governed by Equation (16a.3) and (16a.4) is what makes the transition gradual rather than abrupt. Around the threshold both pathways contribute roughly equal share, and as χ_{n} accumulates further the chunked pathway takes over while the segment-level pathway fades to background. The segment-level pathway is not deleted; its weights remain available, and if χ_{n} decays back below threshold (extended disuse, or after the chunked node is damaged), the system reverts to segment-level execution without losing the underlying capacity. This is what permits recovery patterns observed in clinical cases of focal lesion to chunked motor programs: patients who lose access to overlearned sequences can often re-execute them through slower segment-level construction.

The κ_{trans} parameter controls transition steepness. Low values (around 1) produce a wide transition window where both pathways contribute substantively across many executions, which is biologically realistic but can produce noisy output during the transition. High values (around 10 or more) produce a near-step transition that is less biologically realistic but cleaner in simulation. A starting value around 3 to 5 produces transitions that match the empirical observation that motor skills shift from cortical to subcortical control over weeks of practice rather than abruptly.

17. SMA Initiation Map and Basal Ganglia Speech Loop

Arrow: SSM candidate activations + content buffer state → striatum (caudate, putamen) → GPi/SNr tonic inhibition → MD thalamus → SMA initiation map → gating signal back to SSM. Disinhibitory loop specialized for syllable release rather than general action selection.

$$u_{i}^{\mathrm{str}}(t) \;=\; \mathrm{ReLU}\!\left(\tilde{P}_{i}(t) \cdot b_{k(i)}(t) \right) \tag{17.1}$$

$$\dot{T}_{i}^{\mathrm{GPi}}(t) \;=\; -\alpha_{\mathrm{D1}} \cdot \beta_{\mathrm{DA}} \cdot u_{i}^{\mathrm{str}}(t) + \alpha_{\mathrm{D2}} \cdot (1 - \beta_{\mathrm{DA}}) \cdot u_{i}^{\mathrm{str}}(t) \tag{17.2}$$

$$G_{i}(t) \;=\; \mathbb{1}\!\left[T_{i}^{\mathrm{GPi}}(t) < \theta_{G} \;\wedge\; \sum_{j \neq i} G_{j}(t-1) = 0 \right] \tag{17.3}$$

Variables: $\tilde{P}_{i}(t)$ is the pre-gate SSM candidate activation (the argmax computation from Equation 16.1 before the gate is applied). $b_{k(i)}(t)$ is the matching segment slot in the phonological content buffer of subsection 15b, with $k(i)$ mapping an SSM candidate to its required next segment. $u_{i}^{\mathrm{str}}(t)$ is the striatal activation for SSM candidate i, computed as the conjunction of the SSM candidate activation and the matching buffer content. $T_{i}^{\mathrm{GPi}}(t)$ is the tonic GPi/SNr inhibition on the thalamic relay for candidate i; disinhibitory release happens when this drops below threshold θ_{G}. α_{D1} and α_{D2} are the direct- and indirect-pathway learning rates inherited from Section 7 of the main corpus. \

(\beta_{\mathrm{DA}}\) is the dopamine reward prediction error signal. \(G_{i}(t)\) is the gate signal that the SSM and SMA initiation map both read; the conjunction in Equation (17.3) enforces single-candidate release per cycle.

Timing: The basal ganglia loop runs at the syllable rate, releasing one SSM node per cycle of approximately 150 to 250 ms in fluent speech. The hyperdirect pathway (cortex to subthalamic nucleus to GPi) provides fast broad suppression with latency around 10 to 15 ms, which permits aborting a syllable that has just been released if a higher-level signal demands it (self-monitoring catches an error, social interruption, mid-syllable repair). Tonic inhibition decays back to baseline over roughly 100 ms after a release, preventing two adjacent SSM nodes from firing within the same syllable window.

Mink, J.W. (1996). The basal ganglia: focused selection and inhibition of competing motor programs. Progress in Neurobiology, 50(4), 381–425. DOI: 10.1016/S0301-0082(96)00042-1

Bohland, J.W., Bullock, D. and Guenther, F.H. (2010). Neural representations and mechanisms for the performance of simple speech sequences. Journal of Cognitive Neuroscience, 22(7), 1504–1529. DOI: 10.1162/jocn.2009.21306

Alm, P.A. (2004). Stuttering and the basal ganglia circuits: a critical review of possible relations. Journal of Communication Disorders, 37(4), 325–369. DOI: 10.1016/j.jcomdis.2004.03.001

Engineering Note. This is the speech-specific specialization of Section 7 of the main corpus, not a duplicate. The general basal ganglia loop in Section 7 selects between candidate actions; this loop selects between candidate syllables. Reuse the StriatalGatingSelector from Section 7 with a speech-specific configuration that maps SSM candidates to channels. The biology has one basal ganglia, with different cortical loops projecting through it, and the architecture should preserve that structure or it loses the unified selection that prevents simultaneous speech and non-speech action commitment. The conjunction in Equation (17.1) is the load-bearing detail: striatal activation fires only when the SSM candidate and the matching buffer slot agree, which makes the gate selective rather than just a downstream consequence of SSM competition.

Engineering Note (Stuttering as Loop Pathology). Alm 2004 reviews evidence that developmental stuttering involves dysfunction in this basal ganglia loop, specifically a failure of the timing signal that triggers gate release. In the substrate this corresponds to elevated \(\theta_{G}\) (the gate threshold), which means tonic inhibition has to drop further than normal before the gate opens. The behavioral consequence is delayed or repeated syllable release, which is the acoustic signature of stuttering. The disorder configuration in Appendix G implements this by raising \(\theta_{G}\) and observing the resulting prolongations and repetitions. Dopaminergic medications (D\(_{2}\) antagonists) sometimes worsen stuttering because they shift the balance of Equation (17.2) toward the indirect pathway, which raises \(T_{i}^{\mathrm{GPi}}\) further and makes gate release harder. The substrate predicts this clinical observation as a direct consequence of the equations rather than as a separate model.

18. vPMC to vMC and Corticobulbar Output

Arrow: Speech sound map feedforward command \(\mathbf{m}_{\mathrm{ff}}\) → left ventral motor cortex (vMC) articulator map → corticobulbar projection → brainstem motor nuclei → articulators. The motor command becomes physical movement of the speech articulators after the corticobulbar transmission delay.

\[\mathbf{M}(t) \;=\; \mathbf{M}(t-1) + \alpha_{\mathrm{ff}} \cdot \mathbf{m}_{\mathrm{ff}}(t) \cdot G(t) + \alpha_{\mathrm{fb}} \cdot \mathbf{m}_{\mathrm{fb}}(t) \cdot G(t) \tag{18.1} \] \[\mathbf{Artic}(t)

$$= f_{\mathrm{MAr}}\!\left(\mathbf{M}(t - \tau_{\mathrm{MAr}}) \right) \tag{18.2}$$

$$\tau_{\mathrm{MAr}} \;\approx\; 40 \text{ to } 42 \text{ ms} \tag{18.3}$$

Variables: $\mathbf{M}(t)$ is the 16-dimensional motor cortex position map vector for the eight Maeda articulators (each articulator contributing two dimensions for position and velocity). α_{ff} and α_{fb} are the feedforward and feedback contribution gains, with default values $\alpha_{\mathrm{ff}} \approx 0.9$ and $\alpha_{\mathrm{fb}} \approx 0.1$ in mature speech (the system relies primarily on feedforward). $G(t)$ is the SMA initiation gate from Section 17. f_{MAr} is the motor-cortex-to-articulator transformation that converts cortical commands into physical movement. τ_{MAr} is the corticobulbar transmission delay, empirically measured at approximately 40 to 42 ms from primary motor cortex activation to articulator movement onset.

Timing: The 40 to 42 ms corticobulbar delay is the largest single delay in the production pathway downstream of cortex, and it is the architectural reason a purely feedback-based speech system would be unstable: phoneme transitions in rapid words occur in less than 100 ms, which is faster than two-way feedback could compensate. This is why the brain supplements feedback with feedforward control. The vMC activation begins when the SSM gate fires (around 600 ms post picture onset for a single-syllable picture-naming utterance) and continues throughout the articulation window, with the articulator movement following 40 to 42 ms behind the cortical command.

Guenther, F.H., Ghosh, S.S. and Tourville, J.A. (2006). Neural modeling and imaging of the cortical interactions underlying syllable production. Brain and Language, 96(3), 280–301. DOI: 10.1016/j.bandl.2005.06.001

Maeda, S. (1990). Compensatory articulation during speech: evidence from the analysis and synthesis of vocal tract shapes using an articulatory model. In Hardcastle, W.J. and Marchal, A. (Eds.), Speech Production and Speech Modelling. NATO ASI Series, vol. 55, 131–149. Springer. DOI: 10.1007/978-94-009-2037-8_6

Tourville, J.A., Reilly, K.J. and Guenther, F.H. (2008). Neural mechanisms underlying auditory feedback control of speech. NeuroImage, 39(3), 1429–1443. DOI: 10.1016/j.neuroimage.2007.09.054

Engineering Note. The 16-dimensional articulator vector is not optional. Lower-dimensional motor commands cannot represent the full configuration of the eight Maeda articulators, and higher-dimensional commands lose biological correspondence. Implement $\mathbf{M}$ as exactly 16 dimensions with the named articulator-pair convention from Maeda 1990: jaw position and velocity, tongue body position and velocity, tongue tip position and velocity, lip protrusion position and velocity, lip aperture position and velocity, larynx height position and velocity, and two velum dimensions. This is what makes the Section 21 cerebellar prediction errors anatomically interpretable; if the articulator vector is a black-box latent space, the cerebellum cannot map its predictions onto specific articulators and the substrate loses its ability to model articulator-specific disorders.

Engineering Note (Feedforward/Feedback Balance). The default values $\alpha_{\mathrm{ff}} \approx 0.9$ and $\alpha_{\mathrm{fb}} \approx 0.1$ represent mature speech where the feedforward system has absorbed most corrective information from the feedback loop. Early in learning these values are roughly equal, and the system relies heavily on feedback to drive articulation. The transition is governed by the Equation (16.3) feedforward weight update: as $\mathbf{z}_{\mathrm{PM}}$ absorbs the feedback signal over many production attempts, the feedback contribution shrinks and the feedforward contribution grows. The disorder configuration for childhood apraxia of speech in Appendix G implements Terband et al. 2009 by reducing α_{ff} toward 0.55 and observing the resulting cascade of symptoms (deviant coarticulation, speech sound distortion, searching/groping behavior, increased token-to-token

variability). The Tourville 2008 finding that auditory perturbation increases α_{fb} toward 0.15 is the substrate's compensation mechanism in real time.

19. Auditory and Somatosensory Error Maps (Bilateral pSTG, vSC)

Arrow: Predicted sensory consequences from SSM (top-down predictions) + actual sensory feedback from articulator output (bottom-up signal) → bilateral posterior superior temporal gyrus (auditory error) and ventral somatosensory cortex (somatosensory error) → corrective feedback command $\mathbf{m}_{\mathrm{fb}}$ routed back through right-frontal feedback corrector (Section 20).

$$\mathbf{Au}(t) \;=\; f_{\mathrm{AcAu}}\!\left(\mathbf{Acoust}(t - \tau_{\mathrm{AcAu}}) \right) \tag{19.1}$$

$$\Delta\mathbf{Au}(t) \;=\; \mathbf{Au}(t) - \sum_{i} P_{i}(t - \tau_{\mathrm{PAu}}) \cdot \mathbf{z}_{\mathrm{PAu}}^{(i)}(t) \tag{19.2}$$

$$\Delta\mathbf{S}(t) \;=\; \mathbf{S}(t) - \sum_{i} P_{i}(t - \tau_{\mathrm{PS}}) \cdot \mathbf{z}_{\mathrm{PS}}^{(i)}(t) \tag{19.3}$$

$$\mathbf{m}_{\mathrm{fb}}(t) \;=\; \Delta\mathbf{Au}(t - \tau_{\mathrm{AuM}}) \cdot \mathbf{z}_{\mathrm{AuM}} + \Delta\mathbf{S}(t - \tau_{\mathrm{SM}}) \cdot \mathbf{z}_{\mathrm{SM}} \tag{19.4}$$

Variables: $\mathbf{Au}(t)$ is the auditory state map activity, the post-acoustic-transduction representation that becomes available after the acoustic-to-auditory delay $\tau_{\mathrm{AcAu}} \approx 20$ ms. $\mathbf{S}(t)$ is the somatosensory state map activity. $\mathbf{z}_{\mathrm{PAu}}^{(i)}$ and $\mathbf{z}_{\mathrm{PS}}^{(i)}$ are the predicted auditory and somatosensory targets for SSM node i, retrieved with delays $\tau_{\mathrm{PAu}} \approx 35$ ms and $\tau_{\mathrm{PS}} \approx 20$ ms respectively from premotor cortex. $\Delta\mathbf{Au}(t)$ and $\Delta\mathbf{S}(t)$ are the auditory and somatosensory error maps, the comparator outputs that subtract predicted from actual sensory feedback. $\mathbf{z}_{\mathrm{AuM}}$ and $\mathbf{z}_{\mathrm{SM}}$ are the inverse-Jacobian projections that convert sensory errors back into motor corrections, with delays $\tau_{\mathrm{AuM}} \approx 5$ ms and $\tau_{\mathrm{SM}} \approx 5$ ms. $\mathbf{m}_{\mathrm{fb}}(t)$ is the assembled feedback corrective command.

Timing: Auditory feedback compensation to formant perturbation has measured latency 108 to 165 ms from perturbation onset to articulator response. The component delays decompose as: 20 ms acoustic-to-auditory cortex, plus auditory-error-map computation, plus 5 ms auditory-to-motor cortex transit, plus some processing in motor cortex. The bilateral implementation in posterior superior temporal cortex is essential per Tourville 2008: the auditory error map is bilateral, but the corrective motor commands route preferentially through right-hemisphere frontal regions (Section 20). Somatosensory feedback is faster (around 60 ms total) because the somatosensory pathway has shorter delays.

Tourville, J.A., Reilly, K.J. and Guenther, F.H. (2008). Neural mechanisms underlying auditory feedback control of speech. NeuroImage, 39(3), 1429–1443. DOI: 10.1016/j.neuroimage.2007.09.054

Guenther, F.H., Ghosh, S.S. and Tourville, J.A. (2006). Neural modeling and imaging of the cortical interactions underlying syllable production. Brain and Language, 96(3), 280–301. DOI: 10.1016/j.bandl.2005.06.001

Houde, J.F. and Jordan, M.I. (1998). Sensorimotor adaptation in speech production. Science, 279(5354), 1213–1216. DOI: 10.1126/science.279.5354.1213

Engineering Note. The bilateral implementation of the auditory error map is the empirical commitment that distinguishes this substrate from older speech models. Tourville 2008 demonstrated that the auditory error cells exist in bilateral posterior superior temporal gyrus and planum temporale, and that compensatory articulator movements depend on this bilateral substrate. Implementing the auditory error map only in the left hemisphere reproduces the imaging signature for normal production but loses the right-hemisphere recruitment under perturbation, which is the

diagnostic that confirms the architecture is doing predictive coding rather than just forward production. The somatosensory error map is similarly bilateral. The corrective routing in Section 20 is what introduces the right-hemisphere preference, not the error computation itself.

Engineering Note (Inverse-Jacobian Tuning). The matrices $\mathbf{z}_{\mathrm{AuM}}$ and $\mathbf{z}_{\mathrm{SM}}$ implement the inverse Jacobian that converts sensory errors back into motor corrections. Guenther 2006 treats these as locally linearized inverse mappings tuned during the babbling phase by random articulator perturbation followed by observed sensory consequences. Implement this as a small online regression that updates the inverse-Jacobian matrices when perturbation events occur, with the update rule $\mathbf{z}_{\mathrm{AuM}}(\mathbf{M}) \approx \mathbf{J}(\mathbf{M})^{-1}$ where $\mathbf{J}$ is the local forward Jacobian of the sensory-to-articulator mapping. The classical pitfall here is that the Jacobian is configuration-dependent (different in different articulator postures), so a single fixed inverse cannot capture the full mapping. The substrate handles this by maintaining $\mathbf{z}_{\mathrm{AuM}}$ as a learned function of $\mathbf{M}$ rather than a fixed matrix, but at the cost of much higher parameter count. A reasonable starting implementation freezes the inverse-Jacobian after the babbling phase and accepts the local-linearization error in mature speech, which matches the biology in the sense that adult speech compensation is robust but not infinitely flexible.

19a. Area Spt and the Sensorimotor Translation Interface

Substrate support. No additional equations. This subsection records the role of area Spt in the Sylvian fissure as the acoustic-to-motor translator that complements the arcuate-fasciculus pathway.

Area Spt sits at the parietal-temporal boundary in the Sylvian fissure and provides an acoustic-to-articulatory translation interface that operates in parallel with the arcuate-fasciculus path. While the arcuate carries spelled-out phonological segments from Wernicke's to Broca's for syllabification (Section 14), Spt carries acoustic feature patterns directly to motor planning regions for imitation and online compensation. The two pathways subserve different functions: the arcuate is the standard production-following-perception path used for repetition and dictation, while Spt is the sensorimotor mapping used for imitation of novel sounds and for online articulatory compensation when feedback diverges from prediction.

Empirically, Spt activates during covert speech and during overt repetition, but it does not activate during pure listening or pure self-paced production. This dual activation profile is the signature of a sensorimotor interface rather than a pure perception or pure production substrate. Hickok 2012 argues that Spt is essential for verbal working memory because it provides the mapping from heard speech to the articulatory rehearsal that maintains phonological information in working memory.

The architectural commitment is that Spt operates as a parallel projection from the auditory error map (Section 19) directly to the speech sound map (Section 16), bypassing Wernicke's lexical phonological code retrieval and the arcuate. This is what permits imitation of nonsense syllables that have no lexical entry: the acoustic feature pattern projects through Spt to whatever SSM nodes best match its sensory targets, which activates the corresponding motor commands without ever consulting the lexicon.

Hickok, G. (2012). Computational neuroanatomy of speech production. Nature Reviews Neuroscience, 13(2), 135–145. DOI: 10.1038/nrn3158

Hickok, G. and Poeppel, D. (2007). The cortical organization of speech processing. Nature Reviews Neuroscience, 8(5), 393–402. DOI: 10.1038/nrn2113

Buchsbaum, B.R., Baldo, J., Okada, K., Berman, K.F., Dronkers, N., D'Esposito, M. and Hickok, G. (2011). Conduction aphasia, sensory-motor integration, and phonological short-term memory: an aggregate analysis of lesion and fMRI

data. Brain and Language, 119(3), 119–128. DOI: 10.1016/j.bandl.2010.12.001

Engineering Note. Implement Spt as a thin projection module that maps auditory error map output directly to SSM input, in parallel with the standard arcuate-and-Broca's pathway. The two pathways have different latencies (Spt is faster because it bypasses syllabification) and different output characteristics (Spt produces graded SSM activations matching the acoustic input, while the arcuate path produces categorical SSM activations matching the lexical entry). The substrate should let both pathways run continuously and have the SSM combine their inputs, with the lexical pathway dominating when a clear lexical match exists and the Spt pathway dominating for unfamiliar acoustic input. The combination is automatic if the SSM uses additive integration before the winner-take-all dynamics, which is the default behavior of the Equation (16.1) projection. Do not implement this as a switching mechanism that picks one pathway or the other; the biology runs both in parallel and lets them sum.

20. Right-Frontal Feedback Correction Circuit

Arrow: Bilateral auditory error map $\Delta\mathbf{Au}(t)$ and somatosensory error map $\Delta\mathbf{S}(t)$ → right ventral precentral gyrus and right inferior frontal cortex → corrective motor command routed back to vMC. The compensation pathway is right-lateralized even though normal speech production is left-lateralized.

$$\mathbf{r}_{\mathrm{RH}}(t) \;=\; \sigma\!\left(W_{\mathrm{RH}} \cdot \mathrm{concat}(\Delta\mathbf{Au}(t), \Delta\mathbf{S}(t)) \cdot \beta_{\mathrm{NE}} \right) \tag{20.1}$$

$$s_{\mathrm{dir}}(t) \;=\; \mathrm{sign}\!\left(\mathbf{u}_{F1}^{\top} \cdot \Delta\mathbf{Au}(t) \right) \tag{20.2}$$

$$\tau_{\mathrm{comp}}(t) \;=\; \begin{cases} \tau_{\mathrm{down}} \approx 108 \text{ ms} & \text{if } s_{\mathrm{dir}}(t) < 0 \\ \tau_{\mathrm{up}} \approx 165 \text{ ms} & \text{if } s_{\mathrm{dir}}(t) > 0 \end{cases} \tag{20.3}$$

$$\mathbf{m}_{\mathrm{fb}}^{\mathrm{RH}}(t) \;=\; W_{\mathrm{RH \to vMC}} \cdot \mathbf{r}_{\mathrm{RH}}(t - \tau_{\mathrm{comp}}(t)) \tag{20.4}$$

Variables: $\mathbf{r}_{\mathrm{RH}}(t)$ is the right-frontal feedback corrector activation, a nonlinear function of the bilateral auditory and somatosensory error maps modulated by the norepinephrine arousal scalar β_{NE}. High arousal under perturbation amplifies the corrective response. W_{RH} is the error-to-correction projection in right hemisphere. $W_{\mathrm{RH \to vMC}}$ is the projection back to left vMC for actual articulator correction. $s_{\mathrm{dir}}(t)$ is the perturbation direction sign, computed as the sign of the auditory error projected onto the F1 direction $\mathbf{u}_{F1}$ (the unit vector in auditory feature space corresponding to first-formant frequency). $\tau_{\mathrm{comp}}(t)$ is the total compensation latency from perturbation onset to articulator response, empirically measured at 107.7 ms for shift-down (downward F1 perturbation) and 164.8 ms for shift-up (upward F1 perturbation) per Tourville 2008. The standard deviations are large (around 50 to 60 ms), so the direction-dependent latency is best implemented as a Gaussian random variable with mean τ_{down} or τ_{up} and the empirically measured spread, rather than a deterministic delay.

Timing: The 108 to 165 ms compensation window is the architectural commitment that distinguishes online feedback control from slower offline learning. Tourville 2008 measured mean subject latencies of 107.7 ms in the shift-down condition (standard deviation 61.2 ms) and 164.8 ms in the shift-up condition (standard deviation 43.5 ms), with the asymmetry reflecting either a direction-dependent saliency effect or an underlying circuit-level timing difference. The substrate reproduces the asymmetry through the direction-dependent latency in Equation (20.3) without committing to a specific mechanistic explanation. Within these windows, the right ventral precentral and inferior frontal regions become active in proportion to error magnitude, while the same regions show no differential activation in

normal unperturbed speech. This is the diagnostic that the right-frontal circuit is specifically recruited for error correction, not for general speech production.

Tourville, J.A., Reilly, K.J. and Guenther, F.H. (2008). Neural mechanisms underlying auditory feedback control of speech. NeuroImage, 39(3), 1429–1443. DOI: 10.1016/j.neuroimage.2007.09.054

Toyomura, A., Koyama, S., Miyamaoto, T., Terao, A., Omori, T., Murohashi, H. and Kuriki, S. (2007). Neural correlates of auditory feedback control in human. Neuroscience, 146(2), 499–503. DOI: 10.1016/j.neuroscience.2007.02.023

Max, L., Guenther, F.H., Gracco, V.L., Ghosh, S.S. and Wallace, M.E. (2004). Unstable or insufficiently activated internal models and feedback-biased motor control as sources of dysfluency: a theoretical model of stuttering. Contemporary Issues in Communication Science and Disorders, 31, 105–122. DOI: 10.1044/cicsd_31_S_105

Engineering Note. The right-lateralization of the feedback corrector is the architectural commitment that makes stuttering, apraxia, and right-hemisphere stroke aphasia produce distinct phenotypes in the substrate. Implement W_{RH} and $W_{\mathrm{RH \to vMC}}$ as right-hemisphere-specific weights that exist alongside but separate from the left-hemisphere production pathway weights. Both hemispheres receive the same bilateral error map input, but only the right hemisphere routes that input back to motor cortex through the corrective pathway. A common implementation mistake is to symmetrize the hemispheres and let the substrate learn the lateralization end-to-end, which produces a substrate that violates the empirical lateralization data when ablated. Hard-code the lateralization at initialization and constrain plasticity to operate within rather than across the hemispheric boundary.

Engineering Note (Stuttering as Right-Hemisphere Hyperactivation). Max et al. 2004 propose that developmental stuttering involves elevated reliance on feedback control due to unstable feedforward commands, which produces excess activation in the right-frontal feedback corrector. The substrate predicts this directly: if α_{ff} drops and α_{fb} rises (Section 18 default balance is shifted toward feedback), then the right-frontal corrector receives more input on every utterance and the substrate produces the elevated right-frontal activity that fMRI studies of persons who stutter consistently observe. The disorder configuration in Appendix G implements this by reducing α_{ff} toward 0.7 and observing the substrate's response. Note that this elevated activity is not the cause of stuttering in this model; it is a consequence of the underlying feedforward instability. Treatments that increase right-frontal activity (some forms of fluency training) work by stabilizing the feedforward commands indirectly through the corrective signal, which the substrate also predicts.

21. Cerebellar Speech Specialization (Cb-VI Feedforward, Cb-VIII Feedback)

Arrow: vPMC efference copy → cerebellum lobule VI (feedforward processing) for the production direction; auditory error map → cerebellum lobule VIII (feedback processing) for the corrective direction. Speech-specific specialization of the dual-forward-model cerebellum from Section 8 of the main corpus.

$$\mathrm{pred}_{\mathrm{kin}}^{\mathrm{speech}}(t) \;=\; f_{\mathrm{Cb\text{-}VI}}\!\left(\mathbf{m}_{\mathrm{ff}}(t),\, \mathbf{Artic}(t-1) \right) \tag{21.1}$$

$$\mathrm{pred}_{\mathrm{task}}^{\mathrm{speech}}(t) \;=\; f_{\mathrm{Cb\text{-}VIII}}\!\left(\mathbf{m}_{\mathrm{ff}}(t),\, \mathbf{r}_{\mathrm{aud,target}}(t),\, \mathbf{Au}(t-1) \right) \tag{21.2}$$

$$\mathrm{error}_{\mathrm{kin}}^{\mathrm{speech}}(t) \;=\; \mathbf{Artic}(t) - \mathrm{pred}_{\mathrm{kin}}^{\mathrm{speech}}(t) \tag{21.3}$$

$$\mathrm{error}_{\mathrm{task}}^{\mathrm{speech}}(t) \;=\; \mathbf{r}_{\mathrm{aud,target}}(t) - \mathrm{pred}_{\mathrm{task}}^{\mathrm{speech}}(t) \tag{21.4}$$

Variables: $f_{\mathrm{Cb\text{-}VI}}$ is the kinematic forward model in cerebellar lobule VI, predicting articulator state from motor command and previous articulator state. This is the speech specialization of the kinematic forward model from Section 8 Equation (12a) of the main corpus. $f_{\mathrm{Cb\text{-}VIII}}$ is the task-performance forward model in cerebellar lobule VIII, predicting whether the produced acoustic output will match the intended auditory target $\mathbf{r}_{\mathrm{aud,target}}$. This is the speech specialization of Section 8 Equation (12b). $\mathrm{error}_{\mathrm{kin}}^{\mathrm{speech}}$ routes back to vMC for direct motor correction; $\mathrm{error}_{\mathrm{task}}^{\mathrm{speech}}$ routes back to SSM for syllable selection adjustment, since a kinematically perfect movement that misses the auditory target is a planning failure rather than a motor failure.

Timing: Cerebellar prediction runs in parallel with vMC command emission, with both lobules receiving efference copies simultaneously when the SSM gate fires. The kinematic prediction in Cb-VI is available within roughly 10 ms of efference copy arrival. The task-performance prediction in Cb-VIII has a slightly longer latency because it requires comparing predicted to target auditory outputs, but still completes within the corticobulbar delay so that errors are ready for routing by the time articulator movement begins. The specialization to lobule VI for feedforward and lobule VIII for feedback is empirically supported by Tourville 2008's identification of inferior cerebellum (lobule VIII) activation specifically in the auditory feedback control circuit.

Wolpert, D.M., Miall, R.C. and Kawato, M. (1998). Internal models in the cerebellum. Trends in Cognitive Sciences, 2(9), 338–347. DOI: 10.1016/S0166-2236(98)01312-4

Popa, L.S. and Ebner, T.J. (2019). Cerebellum, predictions and errors. Frontiers in Cellular Neuroscience, 12:524. DOI: 10.3389/fncel.2018.00524

Tourville, J.A., Reilly, K.J. and Guenther, F.H. (2008). Neural mechanisms underlying auditory feedback control of speech. NeuroImage, 39(3), 1429–1443. DOI: 10.1016/j.neuroimage.2007.09.054

Engineering Note. This section is the specialization of Section 8 of the main corpus, not a duplicate cerebellum implementation. Reuse the CerebellarCorrection module from Section 8 with a speech-specific configuration that maps lobule VI to the kinematic forward model and lobule VIII to the task-performance forward model. The biology has one cerebellum with zone-specific function, and the architecture should preserve that structure. The lobule-VI versus lobule-VIII split is what makes the substrate ablation-friendly: a lobule-VI lesion impairs articulator-level kinematic prediction (producing dysarthria-like symptoms), while a lobule-VIII lesion impairs auditory-target tracking (producing more subtle prosodic and intonational deficits). These are different clinical phenotypes and the architecture should be able to produce both.

Engineering Note (Routing of Two Error Signals). The kinematic error $\mathrm{error}_{\mathrm{kin}}^{\mathrm{speech}}$ routes to vMC for direct motor correction because the movement itself was wrong. The task error $\mathrm{error}_{\mathrm{task}}^{\mathrm{speech}}$ routes to SSM (and through SSM upstream to Broca's syllabification if needed) because a kinematically perfect movement that produces the wrong sound is a planning failure. The architectural commitment is that these two errors must not be merged into a single error signal; merging them collapses the dissociation that Popa and Ebner 2019 demonstrated. A common implementation shortcut is to define a single weighted error and route it everywhere, which trains but produces a substrate that cannot reproduce the experimental dissociation between kinematic and task errors. Keep them separate.

22. Anterior Insula: Respiratory-Articulatory Coordination

Arrow: Respiratory state (subglottal pressure, breath phase) + SSM initiation gate → left anterior insula → gain modulation on vMC output. Coordinates the respiratory envelope with articulatory timing so that syllables are produced during exhalation phase with appropriate subglottal pressure.

$$\dot{V}_{\mathrm{lung}}(t) \;=\; \begin{cases} -(V_{\mathrm{lung}}(t) - V_{\mathrm{min}}) / \tau_{\mathrm{exh}} & \text{if exhaling} \\ +(V_{\mathrm{max}} - V_{\mathrm{lung}}(t)) / \tau_{\mathrm{inh}} & \text{if inhaling} \end{cases} \tag{22.1}$$

$$p_{\mathrm{subglott}}(t) \;=\; k_{\mathrm{el}} \cdot (V_{\mathrm{lung}}(t) - V_{\mathrm{rest}}) - R_{\mathrm{lar}} \cdot \dot{V}_{\mathrm{lung}}(t) \tag{22.2}$$

$$\gamma_{\mathrm{ins}}(t) \;=\; \sigma\!\left(W_{\mathrm{ins}} \cdot [\mathbb{1}_{\mathrm{exh}}(t),\, p_{\mathrm{subglott}}(t),\, V_{\mathrm{lung}}(t),\, G(t),\, L_{\mathrm{plan}}(t)] \right) \tag{22.3}$$

$$\mathbf{Artic}_{\mathrm{gated}}(t) \;=\; \gamma_{\mathrm{ins}}(t) \cdot \mathbf{Artic}(t) \tag{22.4}$$

Variables: $V_{\mathrm{lung}}(t)$ is lung volume, the load-bearing physiological variable that drives the rest of the respiratory dynamics. The two-phase update equation captures the asymmetry of speech breathing: exhalation is slow and extended (τ_{exh} on the order of 2 to 5 seconds during connected speech) while inhalation is fast (τ_{inh} on the order of 0.3 to 0.8 seconds). V_{max} and V_{min} are the upper and lower limits of speech-relevant lung volume, typically the inspiratory capacity and the resting expiratory level. V_{rest} is the relaxed-state lung volume below which exhalation requires active expiratory muscle recruitment. $p_{\mathrm{subglott}}(t)$ is subglottal air pressure derived from lung volume through elastic recoil (coefficient k_{el}) plus airflow resistance (R_{lar} is laryngeal resistance, which depends on vocal fold posture). $\mathbb{1}_{\mathrm{exh}}(t)$ is the exhalation indicator (1 during exhalation, 0 during inhalation). $L_{\mathrm{plan}}(t)$ is the planned utterance length signal from prefrontal regions, which the insula uses to determine whether the current V_{lung} is sufficient for the planned utterance. W_{ins} is the insular gating projection. $\gamma_{\mathrm{ins}}(t)$ is the insular gain modulation that suppresses articulation when respiratory readiness is inadequate. $G(t)$ is the SMA initiation gate from Section 17.

Timing: Respiratory coordination operates on the breath cycle of approximately 3 to 5 seconds in connected speech, with the asymmetry of slow exhalation and fast inhalation built into Equation (22.1). This is much slower than the syllable cycle of 150 to 250 ms, so the insular gating signal is a slow envelope that modulates a much faster articulatory stream. Speakers plan utterance length L_{plan} to fit within a single exhalation cycle, and the substrate predicts that speakers with insular damage produce utterances of inappropriate length relative to their respiratory cycle (running out of breath mid-utterance, or initiating utterances during inadequate exhalation), which is a diagnostic signature of apraxia of speech in the Dronkers sense.

Dronkers, N.F. (1996). A new brain region for coordinating speech articulation. Nature, 384(6605), 159–161. DOI: 10.1038/384159a0

Ackermann, H. and Riecker, A. (2010). The contribution(s) of the insula to speech production: a review of the clinical and functional imaging literature. Brain Structure and Function, 214(5-6), 419–433. DOI: 10.1007/s00429-010-0257-x

Hixon, T.J., Goldman, M.D. and Mead, J. (1973). Kinematics of the chest wall during speech production: volume displacements of the rib cage, abdomen, and lung. Journal of Speech and Hearing Research, 16(1), 78–115. DOI: 10.1044/jshr.1601.78

Price, C.J. (2011). A generative model of speech production in Broca's and Wernicke's areas. Frontiers in Psychology, 2:237. DOI: 10.3389/fpsyg.2011.00237

Engineering Note. The anterior insula has its own substrate slot rather than being folded into vMC or SSM because Dronkers 1996 found that insular damage uniquely produced apraxia of speech in her stroke series, with no other lesion site reliably producing the syndrome. This is contested in subsequent literature (some studies implicate vPMC and BA44 in apraxia of speech), but the unique respiratory-articulatory coordination role is empirically robust and is what no other speech region does. Implement the insula as a thin gating module that multiplies the articulator output by a respiratory-derived envelope. Equation (22.1) describes a separate respiratory controller that tracks lung volume independently of speech production; the insula receives the resulting state as a runtime input through W_{ins}.

Engineering Note (Two-Phase Lung Volume Model). The two-phase exponential update in Equation (22.1) replaces a simpler sinusoidal model that earlier drafts of this corpus used. The replacement matters because real breathing is asymmetric: exhalation during connected speech is slow and active (the speaker is using diaphragmatic and abdominal muscle control to extend it for utterance support), while inhalation is fast and largely passive elastic recoil-driven. A pure sinusoid cannot reproduce this asymmetry, which means a sinusoid-based gating function cannot generate the predicted disorder phenotype where insular damage produces inappropriate utterance length. The Hixon, Goldman, and Mead 1973 measurements give the empirical basis for the asymmetric time constants; τ_{exh} of 2 to 5 seconds and τ_{inh} of 0.3 to 0.8 seconds are reasonable starting values that reproduce the kinematics of speech breathing.

The mode-switch between exhaling and inhaling is governed by an upstream respiratory controller that switches based on lung volume (when V_{lung} approaches V_{min}, switch to inhaling; when V_{lung} approaches V_{max} or when no utterance is pending, switch to exhaling). Implement this controller as a separate module from the insula because the two have different roles: the controller manages the breathing cycle whether or not speech is happening, and the insula gates speech output based on the controller's state. Folding them together collapses the architecture's ability to model the case where breathing is disrupted but insular function is intact (sleep apnea, neuromuscular disease) versus the case where insular function is disrupted but breathing is intact (Dronkers-style apraxia of speech).

Engineering Note (Subglottal Pressure Derivation). Equation (22.2) derives subglottal pressure from lung volume rather than treating it as an independently phase-locked variable. The first term $k_{\mathrm{el}} \cdot (V_{\mathrm{lung}} - V_{\mathrm{rest}})$ is elastic recoil pressure: lung tissue and chest wall act like a spring, with pressure proportional to displacement from resting volume. The second term $R_{\mathrm{lar}} \cdot \dot{V}_{\mathrm{lung}}$ is the pressure drop across the larynx during airflow. R_{lar} depends on vocal fold posture and is high during phonation (vocal folds adducted, restricting airflow) and low during voiceless segments. The interaction is what produces the empirically observed pattern where subglottal pressure stays roughly constant during phonation even as lung volume drops, because the speaker compensates through abdominal muscle activity to maintain pressure. A first-pass implementation can hold abdominal compensation constant, but the full prediction of speech-breathing kinematics requires the compensation loop. Add it when the substrate is being validated against respiratory perturbation data.

Engineering Note (Distinct from Apraxia). The disorder configuration in Appendix G distinguishes apraxia of speech (Dronkers-style, insular lesion, breath-articulation coordination failure) from articulatory dysarthria (vMC lesion, articulator weakness) and from BA44 syllabification deficits. The substrate predictions are mechanically distinct: insular damage produces utterances with timing mismatched to respiration; vMC damage produces weak or imprecise articulation regardless of

respiratory state; BA44 damage produces phonological-level errors with intact articulator control. These are genuinely different clinical phenotypes and the substrate's separation of insular respiratory gating from motor and planning regions is what permits them to be modeled as different disorders rather than as gradations of a single deficit. With the lung-volume-and-pressure model in place, the insular disorder configuration can now be tested against actual respiratory perturbation data: ablating W_{ins} should produce utterances initiated at inappropriate V_{lung} levels and continued past the point where subglottal pressure becomes inadequate, which is what apraxia patients show in respiratory-kinematic studies.

23. Internal and External Self-Monitoring Loops

Arrow: Phonological word $\mathbf{w}_{\mathrm{phon}}$ from Broca's syllabification → perception network for internal self-monitoring (catches errors before articulation). Acoustic output from articulators → auditory cortex via external auditory loop → perception network for external self-monitoring (catches errors after articulation). Both loops re-enter the production substrate at the lemma stratum.

$$\mathbf{w}_{\mathrm{monitor}}^{\mathrm{int}}(t) \;=\; W_{\mathrm{phon \to percept}} \cdot \mathbf{w}_{\mathrm{phon}}(t) \tag{23.1}$$

$$\mathbf{w}_{\mathrm{monitor}}^{\mathrm{ext}}(t) \;=\; W_{\mathrm{ac \to percept}} \cdot \mathbf{Acoust}(t - \tau_{\mathrm{ext}}) \tag{23.2}$$

$$e_{\mathrm{monitor}}(t) \;=\; \| \mathbf{w}_{\mathrm{monitor}}(t) - \mathbf{w}_{\mathrm{intended}}(t) \|^{2} \tag{23.3}$$

Variables: $\mathbf{w}_{\mathrm{monitor}}^{\mathrm{int}}$ is the internal monitoring signal, the phonological word output of Broca's syllabification fed back into the perception network without going through articulation. $\mathbf{w}_{\mathrm{monitor}}^{\mathrm{ext}}$ is the external monitoring signal, the acoustic output of the speaker's own voice fed back through the auditory pathway with delay τ_{ext} on the order of 30 to 50 ms (acoustic transduction plus auditory cortex processing). $W_{\mathrm{phon \to percept}}$ is the bypass projection that lets internally-generated phonological words enter the perception substrate without external acoustic feedback. $\mathbf{w}_{\mathrm{intended}}$ is the intended utterance held in working memory, against which both monitoring signals are compared. e_{monitor} is the monitoring error scalar that triggers correction or repair when it exceeds a threshold.

Timing: Internal self-monitoring can begin as early as 355 ms post picture onset, when the first phonological syllable is available from Broca's syllabification (Indefrey and Levelt 2004 calculation). External self-monitoring begins after articulation onset, around 600 ms post picture onset, with the additional 30 to 50 ms acoustic loop delay. Both loops continue throughout articulation and have no clear upper time limit. The internal loop catches errors before they are spoken (the basis for the experimental finding that speakers can begin a self-correction at the same syllable where they detect an error). The external loop catches errors that slip through internal monitoring or arise from articulator perturbation.

Indefrey, P. and Levelt, W.J.M. (2004). The spatial and temporal signatures of word production components. Cognition, 92(1-2), 101–144. DOI: 10.1016/j.cognition.2002.06.001

Levelt, W.J.M. (1983). Monitoring and self-repair in speech. Cognition, 14(1), 41–104. DOI: 10.1016/0010-0277(83)90026-4

Hartsuiker, R.J. and Kolk, H.H.J. (2001). Error monitoring in speech production: a computational test of the perceptual loop theory. Cognitive Psychology, 42(2), 113–157. DOI: 10.1006/cogp.2000.0744

Engineering Note. The internal loop is what makes the substrate able to catch phonological errors before articulation, which is empirically what speakers do (Levelt 1983 documents mid-utterance self-corrections that begin at the same syllable as the detected error). Implement the internal loop as a parallel projection from Broca's syllabification output into the bilateral superior temporal monitoring substrate, bypassing articulation entirely. The implementation pitfall is treating internal monitoring as an after-the-fact check on the articulator output; that conflates internal and external loops and loses the temporal advantage of the internal loop. The two loops use the same downstream perception substrate but receive their inputs from different points in the production pathway, and the architecture should reflect that.

Engineering Note (Self-Monitoring as Perception). McGuire et al. 1996 showed that hearing one's own voice activates the same temporal substrates as hearing another speaker, which means external self-monitoring is not a separate substrate but a re-entry into the standard speech perception pathway. The internal loop is the harder claim: it requires that an internally-generated phonological word can enter the perception pathway without acoustic transduction. The architectural commitment is that $W_{\mathrm{phon \to percept}}$ projects directly from Broca's phonological-word output into Wernicke's lexical phonological code substrate (Section 13), where the representation is recognized as a familiar word and routed to lemma activation in mid-MTG (Section 12). The intended utterance $\mathbf{w}_{\mathrm{intended}}$ is held in working memory at the lemma level, so the comparison happens after the monitoring signal has been perceived as a lemma rather than at the phonological level. This explains why phonemic errors are sometimes corrected and sometimes not: if the phonemic error happens to produce a real word, the internal monitor cannot easily distinguish it from the intended word at the lemma level, and the error slips through.

24. Speech-Specific Neuromodulator Routing

Arrow: Global neuromodulator signals (β_{DA}, β_{NE}, β_{ACh}, β_{5HT}) from main corpus Section 10 → speech-specific receptor routing table → region-specific gain modulations on speech production substrates.

$$\beta^{(r,\mathrm{recep})}_{\mathrm{eff}} \;=\; T_{\mathrm{speech}}[r,\mathrm{recep}] \cdot \beta_{\mathrm{recep}} \tag{24.1}$$

Variables: r indexes speech regions (mid-MTG, Wernicke's, BA44, BA45, vPMC, vMC, anterior insula, right-frontal corrector, cerebellum lobule VI, cerebellum lobule VIII). recep indexes receptor subtypes (D_{1}, D_{2}, α_{1}-NE, α_{2}-NE, M_{1}-ACh, M_{2}-ACh, 5-HT_{1A}, 5-HT_{2A}). T_{speech} is the speech-specific routing table that specifies how each global neuromodulator scalar is amplified or attenuated at each region depending on local receptor density. $\beta^{(r,\mathrm{recep})}_{\mathrm{eff}}$ is the effective modulation applied to region r for receptor type recep, used in the local equations of Sections 11 through 23 wherever a neuromodulator term appears.

Timing: The routing table is configuration, not state. It is loaded at substrate initialization and held constant across the session unless the substrate enters a learning phase that updates it. The neuromodulator scalars themselves update every cycle per Section 10 of the main corpus. The combination of static routing table and dynamic scalars produces region-specific modulation that responds to operating mode in real time.

Avery, M.C. and Krichmar, J.L. (2017). Neuromodulatory systems and their interactions: a review of models, theories, and experiments. Frontiers in Neural Circuits, 11:108. DOI: 10.3389/fncir.2017.00108

Amunts, K. and Zilles, K. (2012). Architecture and organizational principles of Broca's region. Trends in Cognitive Sciences, 16(8), 418–426. DOI: 10.1016/j.tics.2012.06.005

Alm, P.A. (2004). Stuttering and the basal ganglia circuits: a critical review of possible relations. Journal of Communication Disorders, 37(4), 325–369. DOI: 10.1016/j.jcomdis.2004.03.001

Engineering Note. The routing table is a small embedding indexed by (region, receptor) pairs with rows initialized from the receptor density data of Amunts and Zilles 2012 and the neuromodulator review of Avery and Krichmar 2017. The values are scaling factors, typically in the range 0.3 to 2.0, with 1.0 representing the substrate-wide average sensitivity. BA45 has $T[BA45, D_{1}] \approx 1.4$ and $T[BA45, 5HT_{1A}] \approx 1.3$ reflecting its higher D_{1} and serotonergic receptor densities. BA44 has $T[BA44, GABA_{A}] \approx 1.3$ reflecting its higher GABAergic sensitivity. Cerebellum lobule VI and lobule VIII have differential acetylcholine sensitivity reflecting their different roles in feedforward versus feedback circuits. The basal ganglia speech loop of Section 17 has high D_{1} and D_{2} sensitivity which is what makes dopaminergic drugs affect speech production through this substrate. The disorder configurations in Appendix G manipulate specific routing-table entries to reproduce clinical phenotypes (D_{2} antagonist worsening of stuttering, dopaminergic treatment effects on Parkinsonian speech, and so on).

E. Hemispheric Lateralization Commitments

Speech production is largely left-lateralized at the meta-analytic level, with the production pathway from mid-MTG through Wernicke's, the arcuate, Broca's, vPMC, vMC, and the corticobulbar projection all dominated by left-hemisphere activation. The exceptions are architecturally important and they cannot be folded into the left-hemisphere pathway without losing diagnostic capacity for several disorders.

The right hemisphere contributes in the following ways. First, the auditory and somatosensory error maps in posterior superior temporal gyrus and ventral somatosensory cortex are bilateral. Both hemispheres compute the prediction error in parallel, and both feed downstream corrective machinery. Second, the feedback correction pathway is preferentially right-lateralized: the right ventral precentral gyrus and right inferior frontal cortex compute the corrective motor command and route it back to left vMC for articulator adjustment. This lateralization is the substrate of the 108 to 165 ms compensation latency in Section 20. Third, the right cerebellum participates in the dual forward model of Section 21, with lobule VI and lobule VIII roles consistent with their left-cerebellum counterparts but with somewhat different connectivity profiles to right-hemisphere frontal regions.

The lateralization commitment for the substrate is asymmetric weight initialization. The production pathway weights from mid-MTG through vMC are initialized with a 20% leftward bias in cell counts, following Guenther 2006's convention. The error maps are initialized symmetric. The feedback correction pathway is initialized with a 60% rightward bias, reflecting the empirical right-lateralization of compensation-related activity. These are hard biases at initialization that constrain plasticity to operate within rather than across the lateralization profile. The substrate should not be permitted to learn arbitrary lateralization patterns from data; the empirical data is consistent enough across studies that the lateralization is a substrate commitment rather than a learned property.

Tourville, J.A., Reilly, K.J. and Guenther, F.H. (2008). Neural mechanisms underlying auditory feedback control of speech. NeuroImage, 39(3), 1429–1443. DOI: 10.1016/j.neuroimage.2007.09.054

Guenther, F.H., Ghosh, S.S. and Tourville, J.A. (2006). Neural modeling and imaging of the cortical interactions underlying syllable production. Brain and Language, 96(3), 280–301. DOI: 10.1016/j.bandl.2005.06.001

Indefrey, P. and Levelt, W.J.M. (2004). The spatial and temporal signatures of word production components. Cognition, 92(1-2), 101–144. DOI: 10.1016/j.cognition.2002.06.001

Engineering Note. The asymmetric initialization is implemented by giving each substrate region a hemisphere tag and scaling the cell count by the lateralization factor for that region. Plasticity is constrained to operate within hemisphere by masking cross-hemisphere weight updates during training. This is cheap to implement and produces a substrate that reproduces the empirical lateralization patterns from day one rather than waiting for them to emerge from data. The substrate is also more interpretable this way: when something goes wrong with the right-frontal corrector, the diagnostic is straightforward because the corrective pathway has its own substrate slot.

F. Production-Perception Shared Network

Indefrey and Levelt 2004 identify three regions of substrate sharing between word production and word perception: mid-MTG (lemma stratum), Wernicke's posterior superior and middle temporal gyri (lexical phonological code stratum), and Broca's posterior inferior frontal gyrus (involved in both production syllabification and grapheme-to-phoneme conversion during reading). These three sharing points are the substrate of the picture-word interference paradigm, of phonological priming effects, and of the generative-model framing from Price 2011.

The PRAGMI commitment is that the shared substrate uses tied weights, not separate representations that happen to coexist in the same regions. The lemma matrix $W_{\mathrm{C \to L}}$ of Section 12 runs in both production direction (concept to lemma) and comprehension direction (lemma to concept), with the same weight values. The phonological code matrix $W_{\mathrm{L \to P}}$ of Section 13 runs in both production direction (lemma to phonological code) and comprehension direction (phonological code to lemma), again with the same weights. The Broca's syllabification machinery of Section 15 participates in both production (segments to syllables) and comprehension (syllables to segments) of orthographic input during reading, with the same cluster operator running in reverse.

The architectural payoff of tied weights is that semantic interference, phonological facilitation, and self-monitoring fall out of the substrate's natural dynamics rather than requiring additional cross-pathway machinery. A perceived semantically related word activates the distracter lemma, which competes laterally with the target lemma in exactly the same lateral-interference equation that governs production-internal competition. A perceived phonologically related word activates the distracter phonological code, which pre-activates the target's segments because the segmental spell-out matrices are shared. The internal self-monitoring loop reuses the perception substrate to recognize the speaker's own phonological output, which is why monitoring detects the same kinds of errors that auditory comprehension detects when listening to another speaker.

The strong form of the sharing claim is also the easiest to falsify: if the substrate fails to reproduce semantic interference at the predicted SOAs, or if phonological priming fails to show the predicted facilitation pattern, the shared-weights commitment needs to be loosened. The alternative is independent production and perception substrates with explicit cross-pathway coupling, which is more parameters and more machinery for less explanatory power. Start with tied weights and only loosen if the data demand it.

Indefrey, P. and Levelt, W.J.M. (2004). The spatial and temporal signatures of word production components. Cognition, 92(1-2), 101–144. DOI: 10.1016/j.cognition.2002.06.001

Levelt, W.J.M., Roelofs, A. and Meyer, A.S. (1999). A theory of lexical access in speech production. Behavioral and Brain Sciences, 22(1), 1–38. DOI: 10.1017/S0140525X99001776

Price, C.J. (2011). A generative model of speech production in Broca's and Wernicke's areas. Frontiers in Psychology, 2:237. DOI: 10.3389/fpsyg.2011.00237

Engineering Note. PyTorch tied weights are implemented through parameter sharing: declare the matrix once as nn.Parameter and reference it in both the production-direction module and the comprehension-direction module. The transposed access pattern uses .t() in the forward pass without creating new parameters. The implementation pitfall is creating two nn.Linear layers with the same shape and assuming PyTorch will tie them; it will not. Explicit parameter sharing through a single module that holds the weight and exposes both directions is the cleanest pattern, and it makes the shared-weights commitment mechanically enforced rather than relying on training to keep two independent matrices aligned.

G. Disorder Configurations as Falsifiable Predictions

Each disorder configuration is a named parameter bundle that ablates or modifies specific substrate components to reproduce the published phenotype of a clinical disorder. Each configuration is a falsifiable prediction: if the substrate as designed cannot produce the published behavioral signature when configured as specified, then either the substrate is wrong or the published phenotype is wrong, and the discrepancy needs investigation. These are not just labels for failure modes; they are the empirical validation surface for the architecture.

Apraxia of Speech (Dronkers 1996, Terband 2009)

Configuration: reduce α_{ff} to approximately 0.55 in Section 18, holding α_{fb} at 0.45. Ablate or attenuate the anterior insula gating of Section 22 by reducing W_{ins} gain by 50%. Predicted phenotype: deviant coarticulation, speech sound distortion, searching/groping articulatory behavior, increased token-to-token variability, utterance-onset effect (errors clustered at the start of utterances), and respiratory-articulatory mistiming. The substrate produces this phenotype because reduced feedforward gain forces over-reliance on feedback, and attenuated insular gating disrupts the respiratory envelope.

Developmental Stuttering (Alm 2004, Max et al. 2004)

Configuration: raise θ_{G} in Section 17 by approximately 30% above default, making basal ganglia gate release require stronger striatal activation. Optionally reduce α_{ff} toward 0.7 to introduce mild feedforward instability. Predicted phenotype: prolongations and repetitions concentrated at syllable initiation, elevated right-frontal corrector activity (Section 20), and improvement under fluency conditions that bypass the basal ganglia gate (singing, choral reading). The substrate produces this because elevated gate threshold delays SSM release while the syllable representation remains active, producing the characteristic blocked-syllable signature.

Broca's Aphasia (Classical, BA44/BA45 Lesion)

Configuration: ablate the syllabification cluster operator in Section 15 by zeroing W_{44} and W_{45} projections. Leave the lemma stratum (mid-MTG) and phonological code stratum (Wernicke's) intact. Predicted phenotype: agrammatic production with preserved comprehension of single words and short sentences, articulatory effort and phonemic paraphasias, relatively spared lexical access, and difficulty with syntactically complex sentences in comprehension due to the tied-weights commitment that makes Broca's participate in syntactic comprehension. The substrate produces this because lexical and phonological retrieval are upstream of the ablated syllabification.

Wernicke's Aphasia (Classical, Posterior Temporal Lesion)

Configuration: ablate the phonological code retrieval matrix $W_{\mathrm{L \to P}}$ in Section 13. Leave Broca's and the production downstream pathway intact. Predicted phenotype: fluent but semantically empty speech (the production pathway runs but cannot retrieve the right words), severe comprehension deficit (the same matrix runs in comprehension direction), and neologistic paraphasias (the substrate substitutes phonologically possible but lexically incorrect outputs because the lexical phonological code is unavailable). The substrate produces this because the tied-weights commitment makes the same lesion affect both directions.

Conduction Aphasia (Arcuate Lesion)

Configuration: zero the arcuate projection W_{arc} in Section 14 while leaving all cortical substrates intact. Optionally also reduce the Spt projection of subsection 19a to capture cases with combined arcuate and Spt damage. Predicted phenotype: preserved comprehension (Wernicke's intact), preserved spontaneous production (Broca's syllabification intact via direct connections), severe repetition deficit (segmental information cannot transit between Wernicke's and Broca's), and phonemic paraphasias under repetition demand. The substrate produces this because the production pathway has multiple input sources to Broca's syllabification, but only the arcuate carries segments from Wernicke's.

Childhood Apraxia of Speech (Terband 2009)

Configuration: reduce α_{ff} to 0.55 to 0.70 range, simulating the Terband et al. finding that CAS involves degraded feedforward commands. Increase internal neural noise by raising the standard deviation of substrate stochastic terms by approximately 20%. Predicted phenotype: deviant coarticulation, speech sound distortion, searching articulatory behavior, increased token-to-token variability, with particularly elevated errors at utterance onset due to the absence of immediate afferent information at speech start.

Right-Hemisphere Stroke with Speech Compensation Failure

Configuration: ablate the right-frontal corrector of Section 20 by zeroing W_{RH}. Leave the left production pathway intact. Predicted phenotype: preserved fluent speech under unperturbed conditions, but failure to compensate for auditory or articulator perturbation. Speakers with right-hemisphere lesions in the relevant frontal regions show this pattern: their speech is normal until something goes wrong, at which point they cannot adjust. The substrate produces this because the corrective routing is exclusively right-lateralized in this configuration.

Cerebellar Ataxic Dysarthria

Configuration: ablate cerebellar lobule VI in Section 21 by zeroing $f_{\mathrm{Cb-VI}}$. Predicted phenotype: imprecise consonant articulation, irregular articulatory breakdowns, prolonged or distorted vowels, scanning speech with abnormal stress patterns. The substrate produces this because the kinematic forward model is unavailable, leaving vMC without its normal predictive correction.

Dronkers, N.F. (1996). A new brain region for coordinating speech articulation. Nature, 384(6605), 159–161. DOI: 10.1038/384159a0

Terband, H., Maassen, B., Guenther, F.H. and Brumberg, J. (2009). Computational neural modeling of speech motor control in childhood apraxia of speech (CAS). Journal of Speech, Language, and Hearing Research, 52(6), 1595–1609. DOI: 10.1044/1092-4388(2009/07-0283)

Max, L., Guenther, F.H., Gracco, V.L., Ghosh, S.S. and Wallace, M.E. (2004). Unstable or insufficiently activated internal models and feedback-biased motor control as sources of dysfluency: a theoretical model of stuttering.

Contemporary Issues in Communication Science and Disorders, 31, 105–122. DOI: 10.1044/cicsd_31_S_105

Engineering Note. Build the disorder configuration test suite from the start, not as an afterthought. Each configuration should be a named YAML or Python config that the test harness loads and runs, with the test passing if and only if the substrate produces the published behavioral signature. The suite is the empirical validation surface; if a configuration fails, the substrate is wrong somewhere upstream and the failure points to which section of the corpus needs revision. This is also the surface that makes the corpus genuinely scientific rather than just a description: every architectural commitment has a falsification test attached, and the suite either passes or the architecture needs to change. Do not let configurations sit broken; if a configuration cannot be made to pass, the corpus needs to be revised.

H. Speech-Pathway Continuity Frame

Sections 11 through 24 specify the speech motor pathway as a sequence of cortical and subcortical substrates with specific timing windows and connectivity. They do not state the system-level commitment those substrates are in service of, which is that speech identity must not live inside any swappable module. This appendix adds that frame in the same style as Appendix C of the main corpus.

The architectural commitment is that a neuromorphic stack built from the Broca's-corpus substrates must preserve speech identity across hardware upgrades, effector replacements, and acoustic-channel changes. Speech identity is anchored in the lemma stratum (mid-MTG), the lexical phonological code store (Wernicke's), the speech sound map (vPMC), and the chunked motor programs in the syllabary. These are persistent. The articulator effector, the acoustic synthesizer, the auditory transduction front-end, and the corticobulbar projection geometry are mutable infrastructure that can be replaced when hardware improves without erasing what the speaker has learned to say.

Five properties follow from this separation. First, effector swap-compatibility: a hardware upgrade from a low-fidelity vocal tract synthesizer to a high-fidelity one requires retraining the corticobulbar projection (Section 18) and the inverse Jacobian matrices (Section 19), but does not require discarding the lemma store, phonological code store, or syllabary. Second, acoustic-channel independence: replacing the auditory transduction front-end (microphone, direct neural input, cochlear implant simulation) requires recalibrating the auditory feedback loop but does not affect production-side identity. Third, language-acquisition reusability: a speaker who learns a second language acquires new lexical and phonological entries without erasing the first language's entries, because the substrates are additive rather than overwritten. Fourth, dialect and register flexibility: the same lemma store supports multiple phonological realizations through different SSM nodes, which is what permits a speaker to switch register without relearning vocabulary. Fifth, repair after partial damage: the substrate's graceful-degradation property (Spt providing alternative routing when arcuate is damaged, right-frontal compensation when feedforward is impaired) means that recovery from focal lesions can preserve speech identity through redundant pathways even when individual substrates are unavailable.

The integration with the main corpus's Reconamics frame is that speech production is one of the modalities through which the persistent kernel's reconstructed experience reaches the world. The kernel constructs an articulatory trajectory in the same way that imagination constructs a visual scene; the vocal tract describes that trajectory to the listener in the same way that the LLM in the main corpus's Reconamics framing describes reconstructed scenes. The Broca's substrates are the production-side counterpart of the kernel's reconstruction operation, and they inherit the same separation between identity-bearing substrate and replaceable effector that the main corpus commits to.

The test of this commitment is operational. If the substrate can be migrated from one hardware configuration to another, with all peripheral effectors and sensors swapped, and the resulting system speaks in the same voice with the same vocabulary and the same characteristic phonological patterns as before, then Appendix H is honored. If swapping peripherals causes the system to lose speech identity, something in the substrate is absorbing what should be persistent or something persistent is bleeding into what should be replaceable, and the corpus needs revision.

Reconamics: Reconstructive Memory Dynamics for Persistent AI Continuity (Genesis Labs, 2026). Companion paper held in Genesis Labs project knowledge.

Recomics: Modular Cognitive Architecture for Embodied Systems (Genesis Labs, 2026). Companion paper held in Genesis Labs project knowledge.

Engineering Note. The temptation when implementing the Broca's pathway is to let the effector hold state. This shows up as caching articulator parameters in the synthesizer, training the synthesizer end-to-end with the production pathway, or storing learned acoustic calibrations in the auditory transduction front-end. All three collapse the continuity frame. The test is operational: if you can replace the synthesizer or the acoustic transduction with a different implementation of the same interface and the system resumes speaking with its full pre-existing vocabulary and phonological repertoire, the architecture is honoring Appendix H. If anything peripheral holds speech identity, it is absorbing experience it should not own. The cleanest enforcement is to make the substrate components (mid-MTG, Wernicke's, vPMC, the syllabary) the only things serialized between sessions, and to require effectors and sensors to come up cold and recalibrate against the substrate on every fresh boot.